CINEMA THEATRE MANAGEMENT SYSTEM

A PROJECT REPORT

Submitted by

PRANAV SHANMUGAM

(240701395)

in partial fulfilment of the End Semester Examination for the course

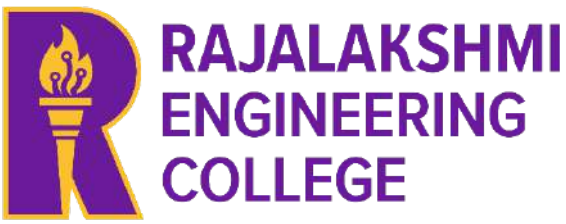
CB23F35 –

LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF ENGINEERING

COMPUTER SCIENCE AND ENGINEERING



RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

Rajalakshmi Nagar, Thandalam, Tamil Nadu – 602 105

NOVEMBER 2026

RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

BONAFIDE CERTIFICATE

Certified that this project report titled "CINEMA THEATRE MANAGEMENT SYSTEM" is the bonafide work of PRANAV SHANMUGAM (240701395), who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027, Semester V, under my supervision.

SIGNATURE

Dr. J. Manoranjini
HEAD OF THE DEPARTMENT
Department of Computer Science and Engineering
Rajalakshmi Engineering College
Thandalam – 602 105

SIGNATURE

Dr. J. Jinu Sophia
Assistant Professor
COURSE FACULTY
Department of Computer Science and Engineering
Rajalakshmi Engineering College

Submitted for the End Semester Examination (Individual Project) held on _____.

INTERNAL EXAMINER

EXTERNAL EXAMINER

DECLARATION

I, PRANAV SHANMUGAM (240701395), hereby declare that the project report titled "CINEMA THEATRE MANAGEMENT SYSTEM", submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its data model, forms, reports, pages, Deluge scripts, workflows and integrations — was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia prediction), code snippet or external resource used in this work has been duly acknowledged in the report.

Place: Chennai
Date: _____

SIGNATURE OF THE STUDENT
PRANAV SHANMUGAM
240701395

ACKNOWLEDGEMENT

I express my sincere gratitude to our Chairman Mr. S. Meganathan, Chairperson Dr. (Mrs.) Thangam Meganathan and Vice-Chairman Mr. M. Abhay Shankar for providing the infrastructure and facilities that made this work possible.

I am grateful to our Principal, Dr. S. N. Murugesan, Rajalakshmi Engineering College, for the constant encouragement extended to students to take up industry-relevant learning.

I thank Dr. J. Manoranjini, Head of the Department, Department of Computer Science and Engineering, for the guidance and support provided throughout the course.

I owe my sincere thanks to my course faculty, Dr. J. Jinu Sophia, Assistant Professor, for the valuable guidance, reviews and feedback that shaped this application and report.

I also acknowledge Zoho Corporation for providing access to the Zoho Creator platform and its learning resources, and I thank my family and friends for their unwavering support.

PRANAV SHANMUGAM

ABSTRACT

Cinema theatres manage several connected activities, including maintaining movie information, scheduling shows, allocating screens, registering customers and processing ticket bookings. Each activity involves records such as movie titles, languages, screen capacities, show timings, ticket prices and customer details. When these records are maintained separately, staff may find it difficult to monitor schedules, verify bookings, coordinate cancellations and retrieve distributor agreement details accurately.

This project presents the Cinema Theatre Management System developed using the Zoho Creator low-code platform. The application stores its data in seven forms — Customer, Movie, Screen, Show, Booking, Cancellation Request and Movie Distribution — connected through lookup fields. Six reports and a dashboard provide organised views of theatre records, schedules and bookings. Form workflows automate booking price calculation, deadline validation, automatic confirmation and cancellation processing. A scheduled workflow is configured to send upcoming-show notifications to the administrator, although it is currently inactive. Deluge scripts provide supporting logic, while Zoho Creator permission profiles control access. The Zoho CRM integration creates corresponding Contact records when customer information is submitted. An AI-based OCR model processes uploaded distributor certificates and extracts eight details: agreement amount, agreement number, distribution territory, distributor name, licence start date, licence end date, movie title and release date.

The application brings customer, movie, screen, show, booking, cancellation and distribution information together within one platform. Booking records are linked to customers and shows, ticket amounts are calculated automatically, and cancellation requests update the corresponding booking status. The project demonstrates the practical use of low-code development, workflow automation, CRM integration and AI-assisted document processing in building a structured cinema theatre management solution.

Keywords: Low-Code Development; Zoho Creator; Cinema Theatre Management; Deluge Scripting; Workflow Automation; Zoho CRM; Optical Character Recognition

PROJECT AT A GLANCE

Attribute	Details
Project Title	Cinema Theatre Management System
Application Domain	Cinema Theatre Operations / Theatre Management
Student Name & Register No.	Pranav Shanmugam – 240701395
Zoho Creator Edition	Trial Edition (Trial currently active)
Application Name (Link Name)	Cinema Theatre Management System (cinema-theatre-management-system)
Published / Live URL	Not specified in the project draft
Demo Credentials (per role)	Administrator, Manager, Staff
Number of Forms / Subforms	7 / 0
Number of Reports (by type)	List: 6; Kanban: 0; Calendar: 0; Pivot/Chart: 0; Spreadsheet: 0; Map: 0; Timeline: 0
Number of Pages / Dashboards	1 Dashboard page
Number of Workflows	Form: 5; Schedule: 1; Approval: 0; Blueprint: 0; Batch: 0
Deluge Functions / Custom Actions	5 embedded Deluge scripts
Integrations	Zoho CRM (Contacts); Zoho Creator AI-based OCR (Movie_Distribution_OCR)
Roles & Profiles Defined	3 Roles: Administrator, Manager and Staff
Environments Used	Zoho Creator Development Workspace
Date of Production Deployment	Not specified in the project draft

TABLE OF CONTENTS

CHAPTER 1 INTRODUCTION 1
1.1 Background 1
1.2 Problem Statement 1
1.3 Objectives 2
1.4 Scope of the Project 2
1.5 Justification for a Low-Code Approach 3
1.6 Organisation of the Report 3
CHAPTER 2 LITERATURE SURVEY AND SYSTEM STUDY 4
2.1 Study of the Existing System 4
2.2 Limitations of the Existing System 5
2.3 Low-Code versus Traditional Development 6
2.4 Literature Survey 7
2.5 Emerging Trends in Low-Code Development 8
2.6 Proposed System 8
CHAPTER 3 REQUIREMENT ANALYSIS 9
3.1 Stakeholders and User Roles 9
3.2 Functional Requirements 9
3.3 Non-Functional Requirements 10
3.4 Hardware and Software Requirements 11
3.5 Use Case Model 11
CHAPTER 4 SYSTEM DESIGN 14
4.1 System Architecture 14
4.2 Data Model 15
4.3 Form Design 16
4.4 Process and Workflow Design 21
4.5 User Interface and Navigation Design 24
4.6 Access Control Design 24
CHAPTER 5 IMPLEMENTATION ON ZOHO CREATOR 26
5.1 Application Setup 26
5.2 Forms and Data Model Implementation 27
5.3 Reports and Dashboard Implementation 27
5.4 Workflow Implementation 28
5.5 Deluge Scripting 29
5.6 Integrations 30
5.7 Users, Roles and Security Configuration 31

CHAPTER 6 TESTING AND DEBUGGING..... 32
6.1 Test Strategy.....32
6.2 Test Cases and Results..... 32
6.3 User Acceptance Testing.....33
6.4 Debugging and Defect Log..... 33
CHAPTER 7 DEPLOYMENT, OPTIMISATION, SECURITY AND GOVERNANCE..... 35
7.1 Deployment.....35
7.2 Performance Optimisation..... 35
7.3 Security..... 36
7.4 Governance and Administration..... 36
CHAPTER 8 RESULTS AND DISCUSSION..... 37
8.1 Achievement of Objectives.....37
8.2 Comparison with the Existing System.....37
8.3 Discussion of Results.....37
8.4 Limitations..... 38
8.5 Low-Code Reflection.....39
CHAPTER 9 CONCLUSION AND FUTURE ENHANCEMENTS.....40
9.1 Conclusion.....40
9.2 Future Enhancements.....40
9.3 Learning Outcomes and Career Pathways..... 40
REFERENCES.....42
APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST..... 43
APPENDIX B – COURSE OUTCOME MAPPING.....49
APPENDIX C – COMPLETE DELUGE SOURCE CODE.....50
C.1 Renewal_Reminder_Function (DS-01).....50
C.2 Budget_Alert_Function (DS-02).....50
C.3 Payment_Notification_Function (DS-03)..... 50
C.4 Cancellation_Processing_Function (DS-04)..... 51
C.5 Auto-Fill Recipient E-mail Field Actions (DS-05, DS-06, DS-07)..... 51
APPENDIX D – USER MANUAL..... 52
D.1 Accessing the Application.....52
D.2 Instructions for the Administrator.....52
D.3 Instructions for the Subscriber..... 53
APPENDIX E – EVALUATION RUBRIC.....54
APPENDIX F – VIVA-VOCE RECORD SHEET..... 56

LIST OF TABLES

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application.....5
Table 2.2: Summary of Literature Survey.....6
Table 3.1: Stakeholders and User Roles.....9
Table 3.2: Functional Requirements.....9
Table 3.3: Non-Functional Requirements.....10
Table 3.4: Hardware and Software Requirements.....10
Table 3.5: Use Case Description – Process a Cancellation Request.....12
Table 4.1: Entities and Relationships.....15
Table 4.2: Field Specification – Customer Form.....15
Table 4.3: Field Specification – Movie Form.....16
Table 4.4: Field Specification – Screen Form.....16
Table 4.5: Field Specification – Show Form.....16
Table 4.6: Field Specification – Booking Form.....17
Table 4.7: Field Specification – Cancellation Request Form.....18
Table 4.8: Field Specification – Movie Distribution Form.....18
Table 4.9: Role–Permission Matrix.....23
Table 5.1: Reports Implemented.....25
Table 5.2: Form and Schedule Workflows.....26
Table 5.3: Blueprints, Approval and Batch Workflows.....27
Table 5.4: Deluge Scripts.....28
Table 6.1: Test Cases and Results.....30
Table 6.2: Acceptance Verification from the Live Dashboard.....31
Table 6.3: Defect Log.....31
Table 7.1: Deployment Checklist.....33
Table 7.2: Performance-Oriented Design Choices.....33
Table 7.3: Security Measures.....34
Table 8.1: Mapping of Objectives to Outcomes.....37
Table 8.2: Existing Manual Process versus the Proposed System.....37
Table A.1: Feature Coverage Checklist.....43
Table B.1: Mapping of Course Outcomes to Report Chapters.....49
Table E.1: Evaluation Rubric for the Individual Project.....54
Table E.2: Performance Level Descriptors.....55
Table F.1: Viva-Voce Questions and Responses.....56
Table F.2: Marks Summary.....56

LIST OF FIGURES

Figure 2.1: As-Is Process Flow of the Existing System.....5
Figure 3.1: Use Case Diagram of the Application..... 12
Figure 4.1: System Architecture of the Application..... 13
Figure 4.2: Entity–Relationship Diagram..... 14
Figure 4.3: Process Flow of the Ticket Booking and Validation..... 19
Figure 4.4: Process Flow of a Cancellation Request.....20
Figure 4.5: Process Flow of Zoho CRM and OCR Integration.....21
Figure 4.6:Navigation Structure and Wireframe of the Dashboard Page.....22
Figure 5.1: Components of the Application in Zoho Creator..... 24
Figure 5.2: Dashboard.....25
Figure 5.3: Block Booking After Deadline Script in the Application IDE.....28
Figure 5.4: Permission Profiles in the Application..... 29
Figure 8.1:Scenario Step 1 – Customer and Ticket Booking Registration..... 36
Figure 8.2:Scenario Step 2 –Automatic Booking Validation & Price Calculation..... 36

LIST OF ABBREVIATIONS

Abbreviation	Expansion
AI	Artificial Intelligence
API	Application Programming Interface
CRM	Customer Relationship Management
CRUD	Create, Read, Update, Delete
ER	Entity–Relationship
FR	Functional Requirement
HTTPS	Hypertext Transfer Protocol Secure
IDE	Integrated Development Environment
INR	Indian Rupee
IST	Indian Standard Time
KPI	Key Performance Indicator
LCAP	Low-Code Application Platform
NFR	Non-Functional Requirement
OCR	Optical Character Recognition
RAM	Random Access Memory
REST	Representational State Transfer
SQL	Structured Query Language
UAT	User Acceptance Testing

CHAPTER 1
INTRODUCTION

1.1 Background

Cinema theatres manage several connected activities every day, including maintaining movie details, scheduling shows, allocating screens, registering customers and processing ticket bookings. Each activity involves information such as movie title, language, duration, screen capacity, show time, ticket price and customer details. When these records are maintained separately, staff may find it difficult to identify available seats, monitor scheduled shows and retrieve booking information accurately.

The main stakeholders in this domain are theatre administrators, who manage movies, screens, shows and booking records, and customers, who select shows and reserve seats. Theatre staff must also coordinate cancellations and ensure that bookings are linked to the correct customer, movie and show. Distribution agreements introduce another administrative requirement, since details such as distributor, territory, licence period and release date must be maintained.

This project addresses these requirements through a Cinema Theatre Management System developed using Zoho Creator. The application maintains structured records for customers, movies, screens, shows, bookings, cancellation requests and movie distribution agreements. It connects related records through lookup relationships and supports the management of theatre operations through forms, reports and workflow automation. These records help the administrator monitor routine activities and locate information when required over time.

1.2 Problem Statement

Theatre operations require a reliable way to manage movie schedules, screen information, customer details and ticket bookings. When these activities are handled through separate manual records, information may become inconsistent, seat availability can be difficult to verify and cancellations may not be reflected correctly. The specific pain points are:

- Movie, screen, show and customer details are maintained separately, making it difficult to obtain a consolidated view of theatre operations.

- Show timings, ticket prices and screen capacities must be checked manually, increasing the possibility of scheduling and booking errors.
- Booking details and seat quantities are not always linked clearly to the corresponding customer and show.
- Cancellations require additional coordination, and the related booking status may not be updated consistently.
- Distribution agreement details, including territory, licence dates and release information, require organised storage and retrieval.

These issues can affect service quality and operational efficiency

1.3 Objectives

- To design forms for maintaining customer, movie, screen, show, booking, cancellation and movie distribution information.
- To establish relationships between customers, bookings, shows, movies and screens using lookup fields.
- To automate booking-related operations, including seat-limit validation, booking status updates and cancellation handling through Zoho Creator workflows.
- To provide reports that help administrators view and manage theatre records, schedules and booking information.
- To use Zoho Creator features, including Deluge scripting and document scanning where applicable, to simplify administrative activities and improve record management.

1.4 Scope of the Project

In scope. Seven forms for customers, movies, screens, shows, bookings, cancellation requests and movie distribution agreements, with reports and lookups. The application supports movie and screen management, show scheduling, customer registration, ticket booking and cancellation handling. Booking records include booking date, seat quantity, status and total price. Screen records maintain screen type and capacity, while show records store date, time, ticket price and status. The Movie Distribution form stores distributor, territory, agreement, licence-period and release details. The

application is accessed through the Zoho Creator web interface, with the administrator managing records.

Out of scope. Live online ticket payments, integration with external ticketing platforms, customer-facing mobile applications, real-time seat selection and automatic synchronisation with distributor databases. These features may be considered as future enhancements.

1.5 Justification for a Low-Code Approach

Low-code platforms allow developers to create business applications using visual form builders, data models and workflow configuration, while using scripts for specialised requirements [1]. Zoho Creator is suitable for this project for five reasons. First, theatre data can be represented through forms and lookup fields, allowing related records to be connected without building a database interface from scratch. Second, reports provide organised views of movies, shows, customers and bookings. Third, workflows can respond to form submissions and record changes, supporting booking validation and cancellation-related updates [2]. Fourth, Deluge scripting can implement conditions and actions that require additional logic. Finally, the application is hosted on the Zoho Creator platform, reducing the need to independently configure servers, databases and deployment infrastructure. This approach makes the system easier to develop, demonstrate and modify within the project environment. It also supports iterative changes during development.

1.6 Organisation of the Report

Chapter 2 studies the existing system and related low-code development concepts. Chapter 3 presents the requirement analysis. Chapter 4 explains the system design. Chapter 5 describes implementation using Zoho Creator. Chapter 6 discusses testing and debugging. Chapter 7 covers deployment, optimisation, security and governance. Chapter 8 presents the results and discussion, and Chapter 9 concludes the report with future enhancements.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

Without a dedicated management application, cinema theatres often maintain their operational information through separate registers, spreadsheets and manually updated records. Movie details are recorded independently, screen availability is checked manually, and show schedules are maintained without a centralised system. When a booking is cancelled, staff must locate the corresponding record and update its status. Distributor agreements are stored as documents, requiring staff to read and enter important information manually. This process makes coordination difficult and increases administrative effort.

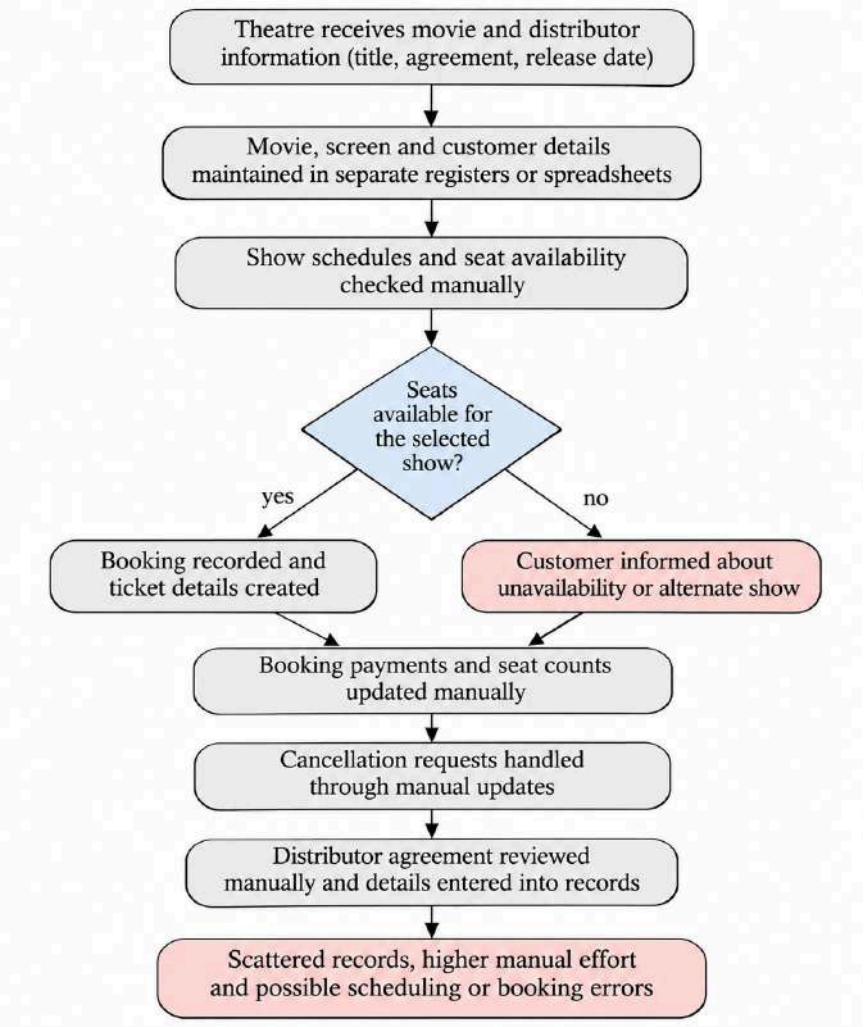


Figure 2.1: As-Is Process Flow of the Existing System

2.2 Limitations of the Existing System

- No centralised information: Movie, screen, show and customer details are maintained separately, making retrieval difficult.
- Manual scheduling: Show timings and screen availability require repeated checking, increasing scheduling errors.
- Booking inconsistencies: Seat quantities, ticket prices and booking statuses require manual verification and updates.
- Unorganised cancellation handling: Cancelled bookings may not be updated consistently, creating confusion in records.
- Manual certificate processing: Distributor agreements must be read manually, and important information has to be entered separately.
- No integrated customer management: Customer information is not automatically synchronised with a dedicated CRM system, making contact management less efficient.

2.3 Low-Code versus Traditional Development

Table 2.1 compares the two development approaches for this application. A conventional implementation would require a database, back-end services, customer management, document-processing functionality, workflow automation and a user interface [3]. On Zoho Creator, these requirements can be addressed through configurable forms, reports, workflows, integrations and AI-assisted features. The effort figures below are estimates for a comparable scope, not measurements.

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development time	Estimated 6–8 weeks for database, backend, CRM, OCR and UI	Estimated 2–3 weeks using forms, workflows and integrations
Skills required	Back-end and front-end programming, SQL, APIs	Business-process modelling and basic Deluge scripting
Hosting & infrastructure	Servers, database and services to be provisioned and maintained	Hosted by Zoho; no servers to manage
Mobile application	Separate responsive or native application build	Phone and tablet layouts available for the page

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Security & access control	Authentication and authorisation coded by hand	Zoho accounts, permission profiles and roles built in
Maintenance & change	Code change, testing and redeployment for every new rule	Change a workflow or script and publish
Customisation limits	Unlimited	Bounded by platform features; extended through Deluge and custom functions
Cost	Developer time plus hosting costs	Platform subscription; trial edition used for this project

2.4 Literature Survey

Table 2.2 summarises eight sources reviewed for this project: research papers on low-code development, Zoho Creator documentation, Deluge scripting and platform capabilities for CRM integration and AI-based document processing.

Table 2.2: Summary of Literature Survey

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	A. C. Bock and U. Frank, 2021 [1]	Low-Code Platform	Defines low-code platforms as environments for visual, model-driven development of applications by users with limited programming skills.	Frames the design approach: forms, workflows and scripts instead of a hand-written code base.
2	Zoho Corporation [2]	Zoho Creator Help Documentation	Describes forms, reports, pages, workflows, blueprints, approvals, AI fields and deployment.	Primary reference for every component built in the application.
3	D. Di Ruscio et al., 2022 [3]	Low-code development and model-driven engineering: two sides of the same coin?	Relates low-code to model-driven engineering and discusses where the two approaches meet.	Supports the use of a data model and process models before configuration.

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
4	M. Waszkowski, 2019 [4]	Low-code platform for automating business processes in manufacturing	Shows that a low-code platform can automate a business process and shorten delivery.	Supports the implementation of booking calculations, cancellation updates and automated theatre operations.
5	A. Sahay et al., 2020 [5]	Supporting the understanding and comparison of low-code development platforms	Proposes a framework to compare low-code platforms based on their features and capabilities.	Supports the selection of Zoho Creator and evaluation of its features for theatre management.
6	Y. Luo et al., 2021 [6]	Characteristics and challenges of low-code development: the practitioners' perspective	Reports practitioner experiences regarding low-code benefits, automation capabilities and platform limitations.	Informs the discussion of customisation, integration capabilities and platform limitations.
7	Zoho Corporation [7]	Deluge Script Reference Guide	Documents Deluge syntax, record operations, CRM tasks, conditional statements and custom functions.	Provides the technical foundation for booking automation, cancellation processing and automatic customer creation in Zoho CRM.
8	Zoho Corporation [8]	Zoho CRM Help Documentation	Explains CRM modules, contact management, record creation and integration with other Zoho applications.	Supports the integration between Zoho Creator and Zoho CRM for automatically creating customer contact records.

2.5 Emerging Trends in Low-Code Development

Four trends influenced the design of this application. **AI-assisted document processing:** applications increasingly use OCR to extract structured information from uploaded documents; this project uses a trained OCR model to identify distributor agreement details. **CRM integration:** business applications connect customer records with external relationship-management platforms, as implemented through Zoho CRM. **Citizen development:** applications can be developed by users through visual configuration rather than extensive programming [1]. **Hyper-automation:** multiple

activities, including booking calculations, cancellation updates and scheduled notifications, can be connected through workflows. **Composable applications:** independent forms, reports, integrations and scripts can work together as a unified application [3].

2.6 Proposed System

The proposed system is a Cinema Theatre Management System developed using Zoho Creator to centralise theatre operations. It addresses the limitations identified in Section 2.2 as follows: Customer, Movie, Screen, Show, Booking, Cancellation Request and Movie Distribution forms maintain structured information; lookup fields connect related records; Deluge workflows calculate booking prices, validate booking deadlines and update cancellation statuses; a scheduled workflow sends upcoming-show notifications; Zoho CRM integration automatically creates customer contact records; and the AI-based OCR model extracts eight important details from uploaded distributor agreements. Reports and a dashboard provide an organised view of theatre activities, improving information management and reducing manual administrative work.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
Administrator	Owner and primary operator of the cinema theatre management application; accesses Zoho Creator.	Manage users, configure application settings, maintain all records and monitor theatre operations.
Manager	Person responsible for supervising cinema theatre activities and reviewing operational information.	Manage movie schedules, review bookings, monitor cancellations and view reports.
Staff / Operator	Theatre employee responsible for handling daily booking and administrative activities.	Register customers, create bookings, update records and process cancellation requests.
Customer / Portal User	Individual whose personal and contact details are maintained in the Customer form.	Provide customer information, request ticket bookings and participate in cancellation processes.

3.2 Functional Requirements

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	The system shall allow administrators to register customers with name, e-mail address and contact details.	Administrator, Staff	High
FR-02	The system shall allow administrators to maintain movie information, including title, language, duration and other details.	Administrator	High
FR-03	The system shall allow screen information to be maintained with screen type and seating capacity.	Administrator	High
FR-04	The system shall allow shows to be scheduled by selecting movies, screens, dates, timings and ticket prices.	Administrator, Manager	High
FR-05	The system shall allow staff to create ticket bookings by selecting customers, shows and ticket quantities.	Staff / Operator	High

FR ID	Requirement	Role(s)	Priority
FR-06	The system shall automatically calculate the total booking amount using ticket quantity and ticket price.	System	High
FR-07	The system shall restrict individual bookings to a maximum of five tickets.	System	High
FR-08	The system shall validate booking conditions and display appropriate messages when booking limits or availability conditions are not satisfied.	System	High
FR-09	The system shall allow cancellation requests to be recorded with the corresponding booking information.	Staff / Operator	High
FR-10	The system shall automatically update the linked booking status to Cancelled when a cancellation request is submitted.	System	High
FR-11	The system shall maintain movie distribution information, including distributor details, agreement number, territory and licence period.	Administrator	High
FR-12	The system shall accept uploaded distributor certificates and process them using the configured AI-based OCR model.	Administrator, AI OCR Model	High
FR-13	The system shall extract agreement amount, agreement number, distribution territory, distributor name, licence start date, licence end date, movie title and release date from uploaded documents.	AI OCR Model	High
FR-14	The system shall automatically create corresponding Contact records in Zoho CRM when customer information is submitted through Zoho Creator.	System, Zoho CRM	High
FR-15	The system shall provide reports for viewing and managing customer, movie, screen, show and booking information.	Administrator, Manager	Medium

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Performance	Reports shall retrieve and display stored customer, movie, show and booking information without unnecessary manual calculations.
NFR-02	Security	Only authorised Zoho users shall access application records through the configured access permissions.

NFR ID	Category	Requirement
NFR-03	Usability	The application shall provide an understandable interface through Zoho Creator forms and reports for managing theatre operations.
NFR-04	Availability	The application shall be hosted on Zoho Creator's cloud platform and accessed through an internet-connected web browser.
NFR-05	Maintainability	Application forms, fields, workflows and Deluge scripts shall be configurable to accommodate changes in theatre operations.
NFR-06	Auditability	Customer, movie, show, booking and cancellation records shall be maintained for administrative review and reference.

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator (Trial Edition)
Scripting language	Deluge (Data Enriched Language for the Universal Grid Environment)
Browser	Google Chrome / Microsoft Edge / Mozilla Firefox (latest version)
API testing tool	Not separately required for the configured Zoho CRM integration
Other Zoho services	Zoho CRM and Zoho Creator AI/OCR integration
Mobile	Zoho Creator mobile application (Android / iOS), if used
Client hardware	Intel Core i3/i5 or above, 8 GB RAM, internet connection

3.5 Use Case Model

Figure 3.1 shows the use cases of the system. The Administrator manages providers, subscriptions, payments, budgets and cancellation requests and views the dashboard. The Scheduler and workflow engine send the renewal reminders and the budget and payment alerts, which are received by the Subscriber.

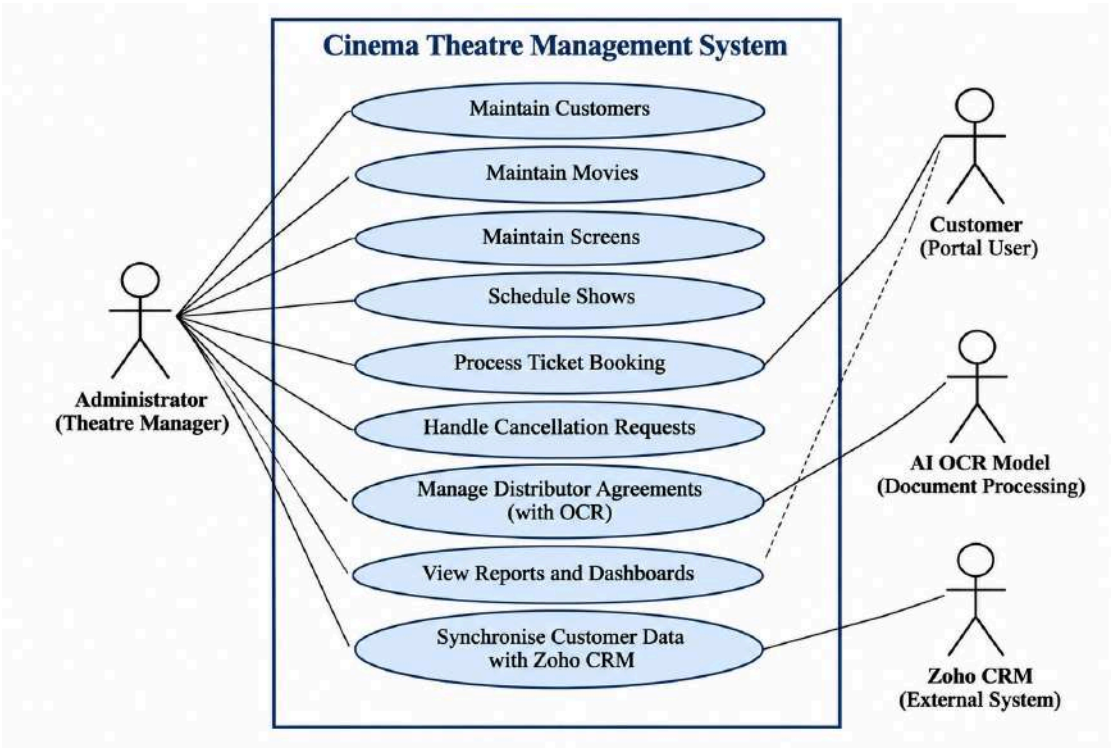


Figure 3.1: Use Case Diagram of the Application

Table 3.5: Use Case Description – Process a Cancellation Request

Field	Description
Use Case ID	UC-01
Use Case Name	Process a Ticket Booking
Actor(s)	Staff / Operator; Workflow Engine
Pre-condition	Customer and show details must be available in the application.
Main Flow	1. The Staff opens the Booking form and selects the customer and show. 2. The required ticket quantity is entered. 3. The system validates the ticket quantity against the maximum limit of five tickets. 4. The system automatically calculates the total booking amount using the ticket quantity and ticket price. 5. The booking form is submitted and the record is stored.
Alternate Flow	If the ticket quantity exceeds five or the booking conditions are not satisfied, the system displays an appropriate validation message.
Post-condition	The booking record is stored with the selected customer, show, ticket quantity and calculated amount.

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

Figure 4.1 shows the layered architecture. Users access the application through a web browser. The application layer consists of the dashboard page, seven forms and their corresponding reports. The logic and automation layer contains Deluge scripts and workflows that handle booking calculations, booking deadline validation, automatic booking confirmation, cancellation processing and scheduled notifications. The data and services layer consists of the Zoho Creator data store, Zoho CRM integration for customer contact management and the AI-based OCR model for distributor certificate processing.

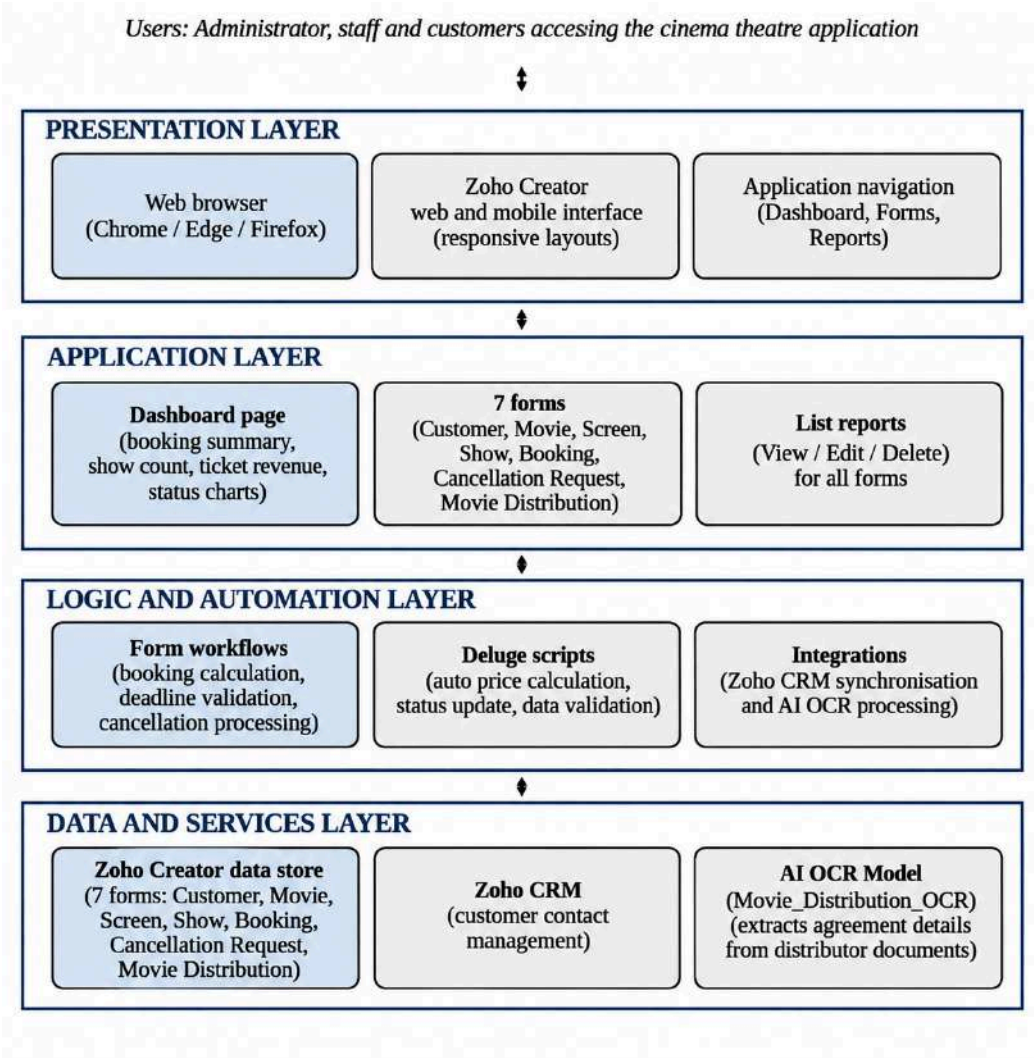


Figure 4.1: System Architecture of the Application

4.2 Data Model

Figure 4.2 presents the entity–relationship diagram. Customer, Movie and Screen are master entities used by the operational records. Show connects Movie and Screen through lookup fields, establishing relationships between scheduled movies and their allocated screens. Booking contains lookup fields for Customer and Show, connecting each booking to the corresponding customer and scheduled show. Cancellation Request contains lookup fields for Booking and Customer, allowing cancellation information to be associated with the original booking. Movie Distribution maintains uploaded distributor agreements and the information extracted through OCR. Table 4.1 lists the entities and their relationships.

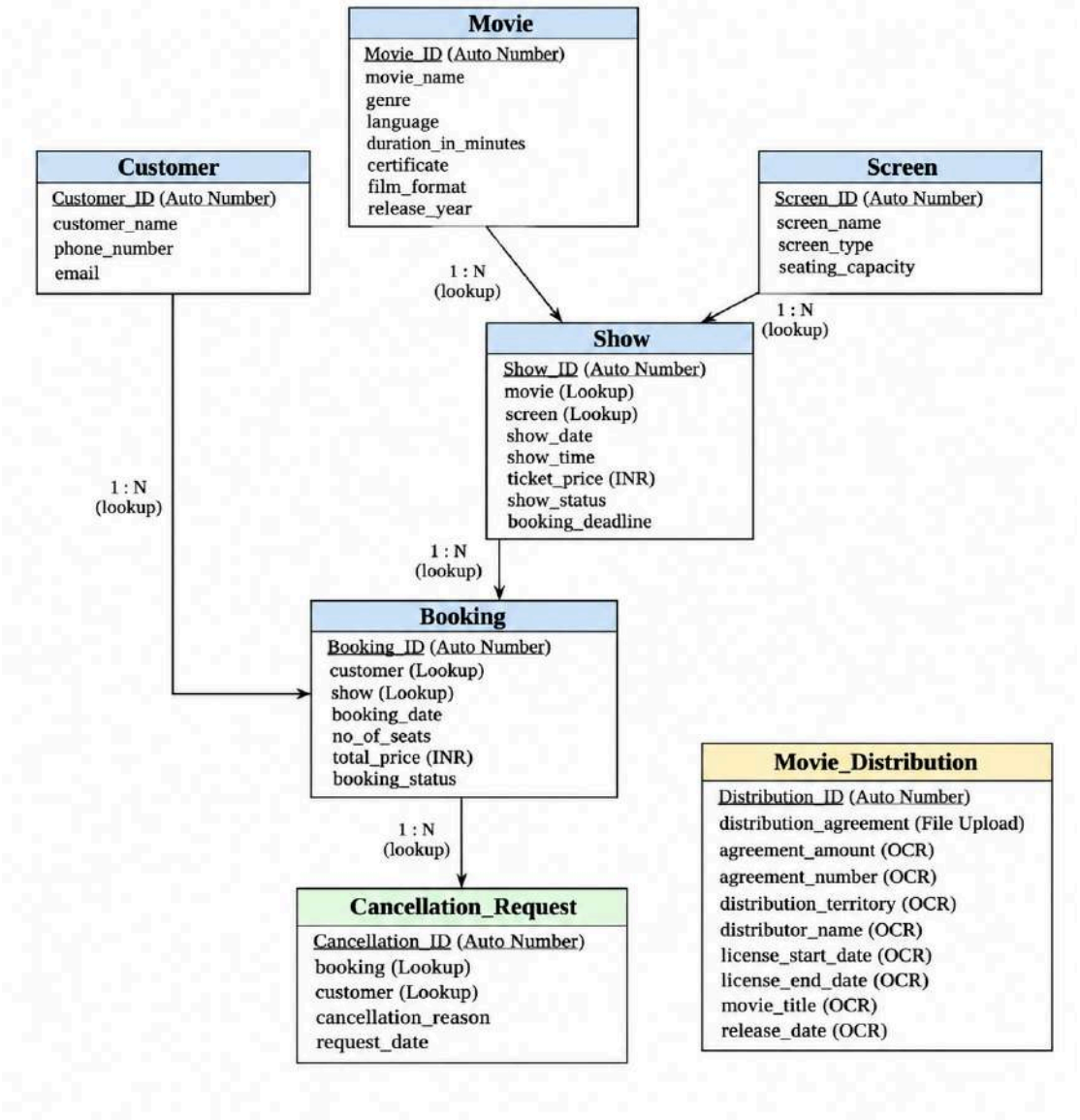


Figure 4.2: Entity–Relationship Diagram

Table 4.1: Entities and Relationships

Entity (Form)	Key Fields	Related To	Relationship Type	Implemented Using
Customer	Name, Phone Number, Email	Booking, Cancellation Request	One-to-Many	Lookup fields in related forms
Movie	Movie Name, Genre, Language, Duration, Certificate, Film Format, Release Year	Show	One-to-Many	Lookup field in Show
Screen	Screen Name, Screen Type, Seating Capacity	Show	One-to-Many	Lookup field in Show
Show	Movie, Screen, Show Date, Show Time, Ticket Price, Show Status, Booking Deadline	Booking	One-to-Many	Lookup field in Booking
Booking	Customer, Show, Booking Date, No. of Seats, Total Price, Booking Status	Cancellation Request	One-to-Many	Lookup field in Cancellation Request
Cancellation Request	Booking, Customer, Cancellation Reason	Booking, Customer	Many-to-One	Lookup fields

4.3 Form Design

Tables 4.2 to 4.8 give the field specification of each form, taken from the form definitions in the Application IDE. The forms use different field types, including text, number, date, time, currency, dropdown, lookup and OCR-enabled fields. Lookup fields establish relationships between customers, movies, screens, shows and bookings. The Movie Distribution form uses the configured AI OCR model to extract distributor agreement details from uploaded documents.

Table 4.2: Field Specification – Customer Form

Field Label	Link Name	Field Type	Required	Validation / Default	Remarks
Name	name	Name	No	—	Customer’s full name
Phone Number	phone_number	Phone	No	—	Customer’s contact number

Field Label	Link Name	Field Type	Required	Validation / Default	Remarks
Email	email	Email	No	—	Customer's email address

Table 4.3: Field Specification – Movie Form

Field Label	Link Name	Field Type	Required	Validation / Default	Remarks
Movie Name	movie_name	Single Line	No	—	Name of the movie
Genre	genre	Drop Down	No	Configured choices	Movie category
Language	language	Drop Down	No	Configured choices	Movie language
Duration in Minutes	duration_in_minutes	Number	No	—	Movie duration
Certificate	certificate	Drop Down	No	Configured choices	Movie certification
Film Format	film_format	Drop Down	No	Configured choices	Cinema film format
Release Year	release_year	Number	No	—	Year of release

Table 4.4: Field Specification – Screen Form

Field Label	Link Name	Field Type	Required	Validation / Default	Remarks
Screen Name	screen_name	Single Line	No	—	Screen identification
Screen Type	screen_type	Drop Down	No	Configured choices	Type of theatre screen
Seating Capacity	seating_capacity	Number	No	—	Maximum seating capacity

Table 4.5: Field Specification – Show Form

Field Label	Link Name	Field Type	Required	Validation / Default	Remarks
Movie	movie	Lookup	No	From Movie	Selects the movie
Screen	screen	Lookup	No	From Screen	Selects the allocated screen
Show Date	show_date	Date	No	—	Scheduled show date
Show Time	show_time	Time	No	—	Scheduled show timing
Ticket Price	ticket_price	Currency (INR)	No	—	Price per ticket
Show Status	show_status	Radio Buttons	No	Configured choices	Current show status
Booking Deadline	booking_deadline	Date-Time	No	—	Last permitted booking date and time

Table 4.6: Field Specification – Booking Form

Field Label	Link Name	Field Type	Required	Validation / Default	Remarks
Customer	customer	Lookup	No	From Customer	Selects the customer
Show	show	Lookup	No	From Show	Selects the scheduled show
Booking Date	booking_date	Date	No	—	Date of booking
No. of Seats	no_of_seats	Number	No	—	Number of tickets booked
Total Price	total_price	Currency (INR)	No	Calculated through workflow	Automatically calculated using ticket quantity and ticket price
Booking Status	booking_status	Drop Down	No	Automatically set to Confirmed	Maintains booking status

Table 4.7: Field Specification – Cancellation Request Form

Field Label	Link Name	Field Type	Required	Validation / Default	Remarks
Booking	booking	Lookup	No	From Booking	Selects the original booking
Cancellation Reason	cancellation_reason	Multi Line	No	—	Reason for cancellation
Customer	customer	Lookup	No	From Customer	Identifies the customer

Table 4.8: Field Specification – Movie Distribution Form

Field Label	Link Name	Field Type	Required	Validation / Default	Remarks
Distribution Agreement	Distribution_Agreement	Image / File Upload	No	Document upload	Uploaded distributor certificate
Agreement Amount	Agreement_Amount1	Number (OCR)	No	AI extraction	Extracted agreement amount
Agreement Number	Agreement_Number1	Single Line (OCR)	No	AI extraction	Extracted agreement number
Distribution Territory	Distribution_Territory1	Single Line (OCR)	No	AI extraction	Extracted distribution territory
Distributor Name	Distributor_Name1	Single Line (OCR)	No	AI extraction	Extracted distributor name
Licence End Date	License_End_Date1	Date (OCR)	No	AI extraction	Extracted licence end date
Licence Start Date	License_Start_Date1	Date (OCR)	No	AI extraction	Extracted licence start date
Movie Title	Movie_Title1	Single Line (OCR)	No	AI extraction	Extracted movie title
Release Date	Release_Date1	Date (OCR)	No	AI extraction	Extracted movie release date

4.4 Process and Workflow Design

Booking management: Figure 4.3 shows the booking process. When a booking is created, the Booking form uses the selected show to retrieve the ticket price. The `auto_calculate_booking_price` workflow calculates the total amount by multiplying the number of seats by the ticket price. The `block_booking_after_deadline` workflow checks the selected show's booking deadline and displays a validation message if the booking date exceeds the permitted deadline. The `auto_set_booking_status_confirmed` workflow assigns the Confirmed status to newly created bookings.

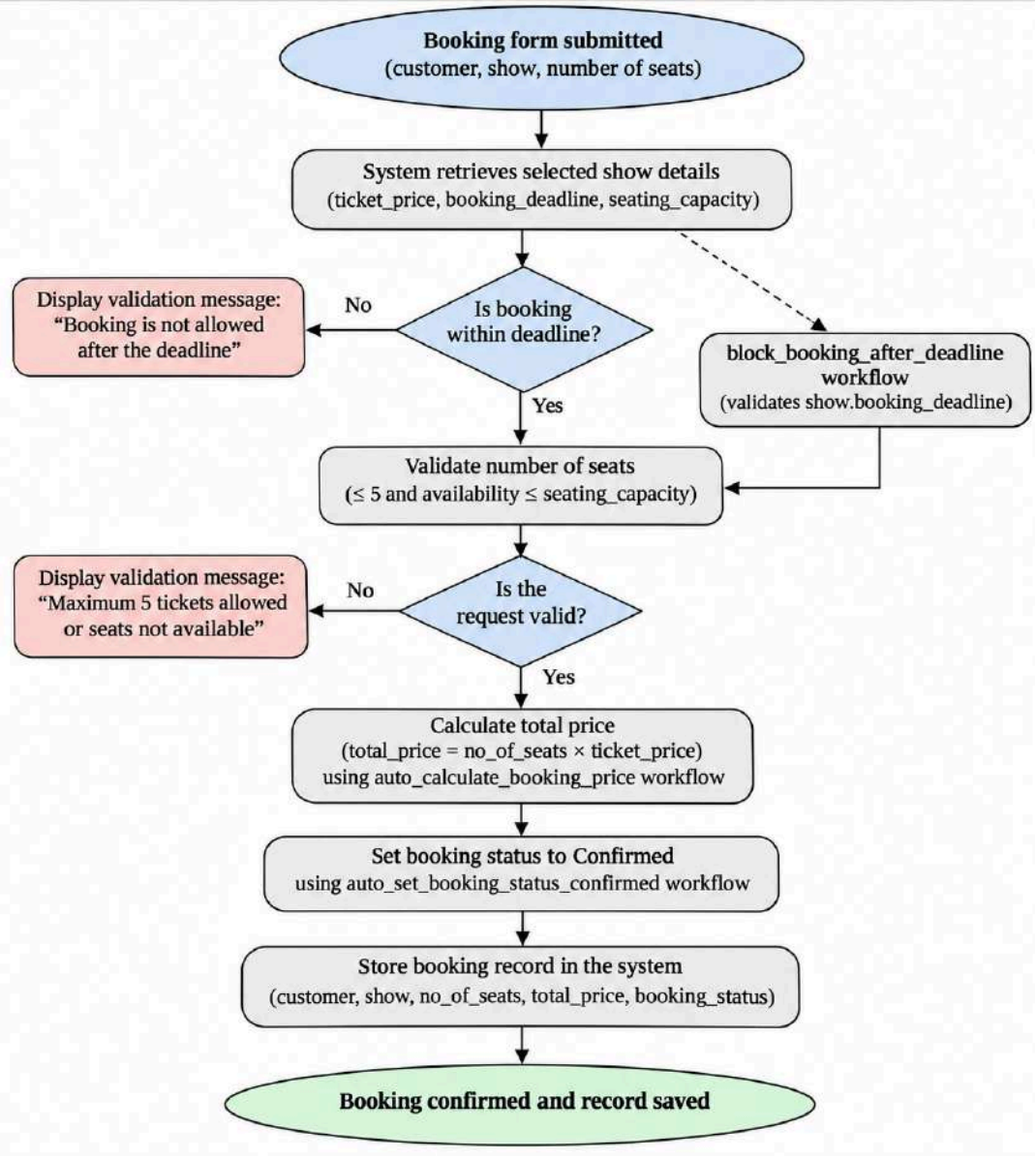


Figure 4.3: Process Flow of the Ticket Booking and Validation

Cancellation handling: Figure 4.4 shows how a cancellation request is processed. When a cancellation request is submitted, the `cancel_booking_on_submission` workflow retrieves the linked booking record and updates its booking status to Cancelled. A confirmation message is displayed after the update. This ensures that the original booking record reflects the cancellation request.

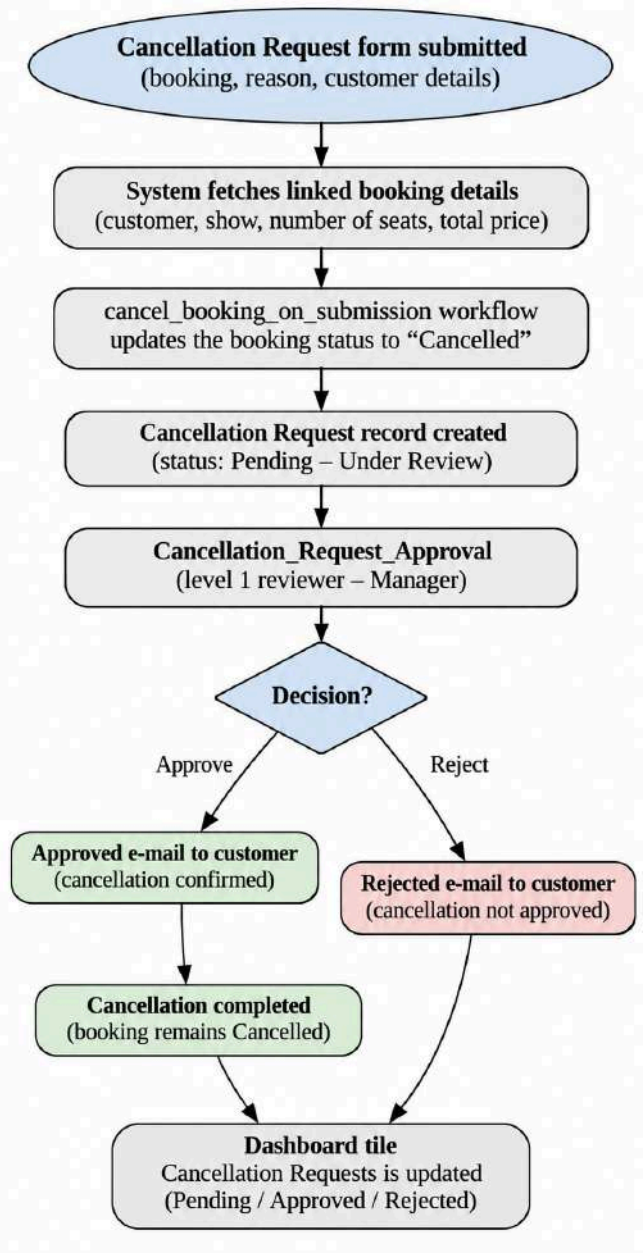


Figure 4.4: Process Flow of a Cancellation Request

CRM and OCR integration: When a Customer record is submitted, the `Sync_Customer_to_Zoho_CRM` workflow creates a corresponding Contact record in Zoho CRM using the customer's name, phone number and e-mail address. The Movie Distribution form uses the `Movie_Distribution_OCR` model to extract eight fields from uploaded distributor agreements, including distributor name, agreement number, territory, amount and licence dates.

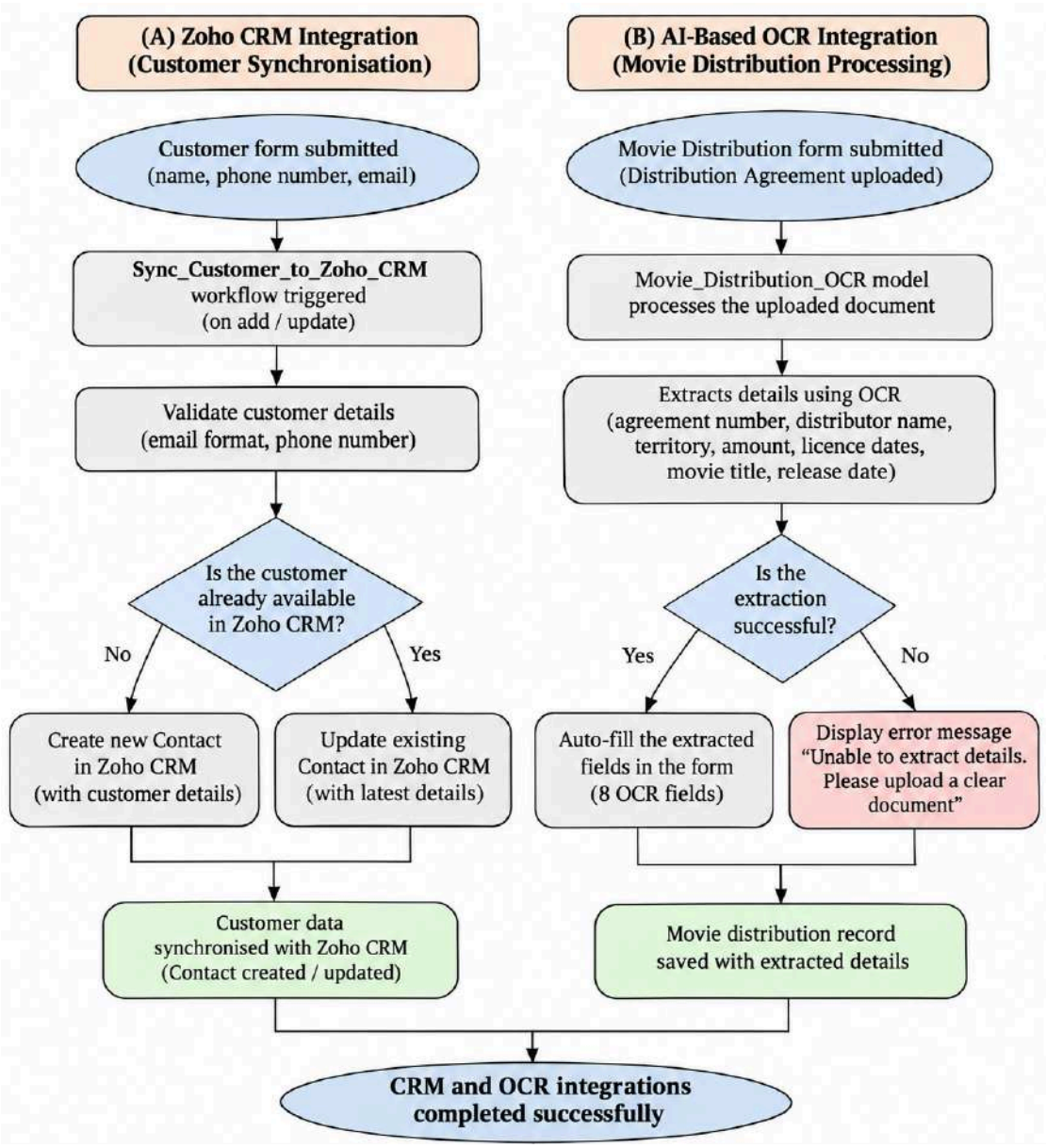


Figure 4.5: Process Flow of Zoho CRM and AI-Based OCR Integration

4.5 User Interface and Navigation Design

The application navigation provides access to Dashboard, Customer, Movie, Screen, Show, Booking, Cancellation Request and Movie Distribution sections. Each data section contains its respective form and list report. The Dashboard page acts as the main landing page, presenting ticket price information, show counts, booking counts and booking status summaries. It also provides quick-access buttons for adding new shows, bookings, screens and customers, together with charts and reports for monitoring theatre operations (Figure 4.6).

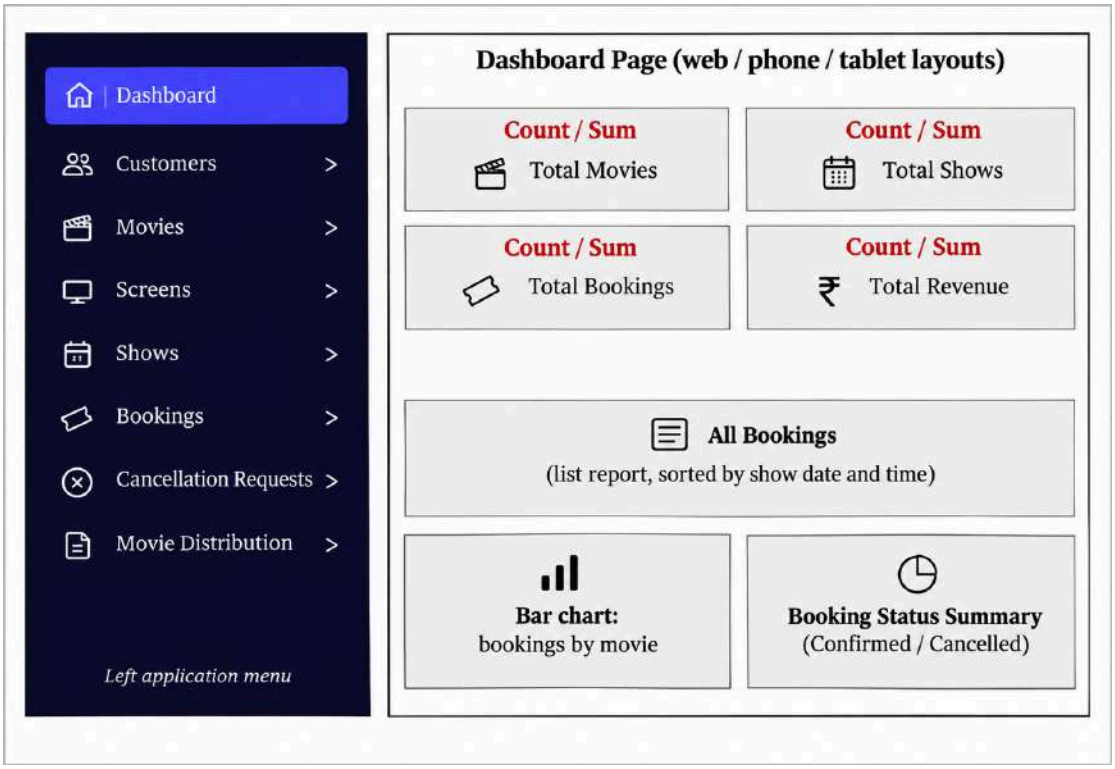


Figure 4.6: Navigation Structure and Wireframe of the Dashboard Page

4.6 Access Control Design

Access is designed around the application owner, who acts as the Administrator, and the configured permission profiles in Zoho Creator. The application includes profiles such as Ticketing Agent, Guest, Show Scheduler, Content Administrator and Cinema Manager, along with Administrator and other system profiles. These profiles provide different levels of access to forms and reports based on their assigned permissions. Table 4.9 maps the major application components to the relevant access roles.

Table 4.9: Role–Permission Matrix

Component	Administrator	Manager	Staff
Form: Customer	A E V D	V	A E V
Form: Movie	A E V D	V	—
Form: Screen	A E V D	V	—
Form: Show	A E V D	E V	V
Form: Booking	A E V D	V	A V
Form: Cancellation Request	A E V D	E V	A V
Form: Movie Distribution	A E V D	V	—
Report: All Customers	V	V	V
Report: All Movies	V	V	—
Report: All Shows	V	V	V
Report: All Bookings	V	V	V (own)
Page: Manager Dashboard	V	V	—

CHAPTER 5
IMPLEMENTATION ON ZOHO CREATOR

5.1 Application Setup

The application “Cinema Theatre Management System” was created in Zoho Creator with the link name `cinema-theatre-management-system`. The application was developed to manage cinema operations, including customer registration, movie management, screen allocation, show scheduling, ticket booking, cancellation handling and distributor agreement management. The application contains seven forms, six reports and one Dashboard page. Figure 5.1 shows the component list of the application as displayed in Zoho Creator, and Figure 5.2 shows the Booking form in the Application IDE.

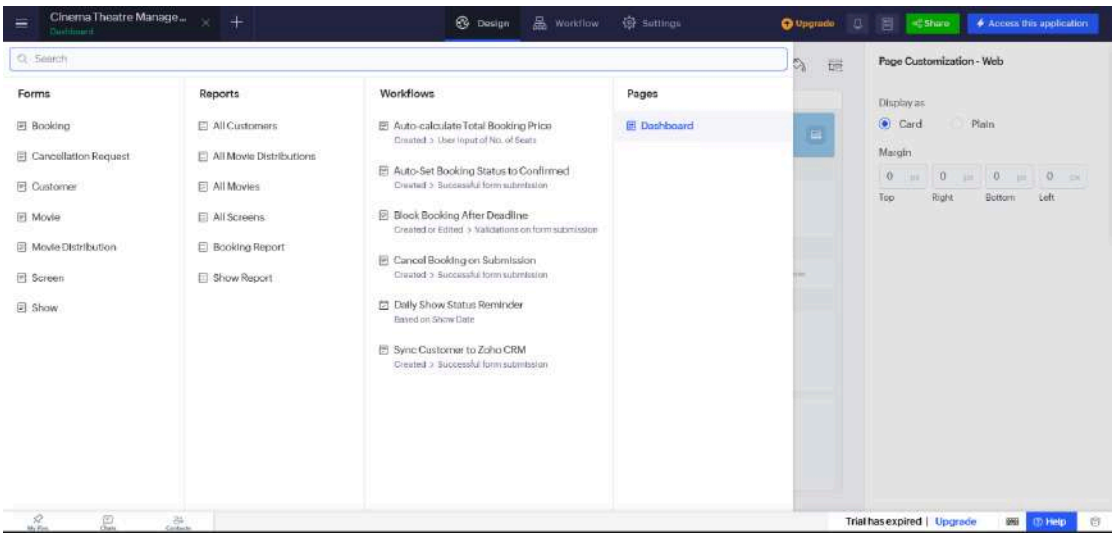


Figure 5.1: Components of the Application in Zoho Creator

5.2 Forms and Data Model Implementation

Seven forms were created (Section 4.3). These forms maintain customer, movie, screen, show, booking, cancellation and distributor agreement information. Autonumber fields generate unique record identifiers, while lookup fields connect related records. The Show form links Movie and Screen, while Booking links Customer and Show. Cancellation Request links to Booking and Customer. The Movie Distribution form uses the configured `Movie_Distribution_OCR` model to extract eight agreement details from uploaded documents. These fields include

agreement amount, agreement number, distribution territory, distributor name, licence start date, licence end date, movie title and release date.

5.3 Reports and Dashboard Implementation

Each form has an associated list report that provides an organised view of stored records. The reports display relevant fields and support record management through the available actions. Table 5.1 lists the reports.

Table 5.1: Reports Implemented

Report	Based On	Type	Notes
Show Report	Show	List	Displays movie, screen, show date, timing, ticket price and booking deadline; sorted by show date
Booking Report	Booking	List	Displays customer, show, booking date, seat quantity, total price, booking status and movie
All Screens	Screen	List	Displays screen name, screen type and seating capacity
All Customers	Customer	List	Displays customer name, phone number and e-mail address
All Movies	Movie	List	Displays movie name, genre, language, duration, certificate, film format and release year
All Movie Distributions	Movie Distribution	List	Displays uploaded agreements and extracted distributor information

The Dashboard page provides an overview of theatre operations through data summaries, charts and quick links. It contains four main information panels: Total Ticket Prices, Count of Show, Count of Booking and Booking Status. The Booking Status section displays Confirmed, Cancelled and Total booking counts. The page also includes a bar chart showing Booking Status by Total Price, another chart showing Distribution by Movie, and a Show Report. Quick links provide access to the Show, Booking, Screen and Customer forms. Figures 5.3 and 5.4 show the dashboard and its chart components.

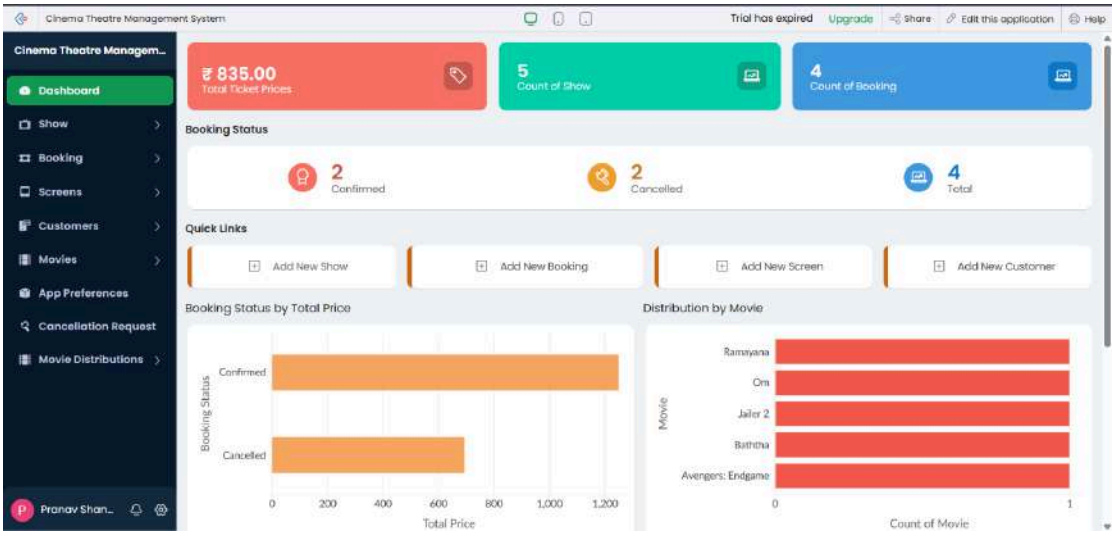


Figure 5.2: Dashboard

5.4 Workflow Implementation

Five form workflows and one scheduled workflow are configured to automate the application. These workflows handle booking price calculation, booking deadline validation, automatic booking confirmation, customer synchronisation with Zoho CRM and cancellation processing. A scheduled workflow is also configured to identify upcoming shows and send an e-mail notification to the administrator. The scheduled workflow is currently inactive. Tables 5.2 and 5.3 describe the configured workflows and other automation components.

Table 5.2: Form and Schedule Workflows

Workflow	Type	Trigger / Condition	Action
Auto_calculate_booking_price	Form	Number of seats entered in Booking form	Retrieves ticket price from selected show and calculates total booking price
Block_Booking_After_Deadline	Form	Booking form validation on add or edit	Checks booking date against show booking deadline and prevents invalid submission
Auto_Set_Booking_Status_Confirmed	Form	New booking successfully submitted	Sets booking status to Confirmed and displays a success message

Workflow	Type	Trigger / Condition	Action
Sync_Customer_to_Zoho_CRM	Form	Customer record added successfully	Creates a corresponding Contact record in Zoho CRM using customer name, phone number and e-mail
Cancel_Booking_on_Submission	Form	Cancellation Request submitted successfully	Retrieves the linked booking and updates its status to Cancelled
Daily_Show_Status_Reminder	Schedule	Configured for the Show form; scheduled time 10:00 AM Asia/Kolkata	Fetches shows scheduled for the following day and prepares an e-mail notification for the administrator; currently inactive

Table 5.3: Blueprints, Approval and Batch Workflows

Workflow	Type	Stages / Behaviour
Booking Approval	Blueprint	Not implemented
Cancellation Approval	Approval	Not implemented; cancellation is processed automatically through a form workflow
Booking Processing	Batch	Not implemented
Show Scheduling	Batch	Not implemented

5.5 Deluge Scripting

Five embedded Deluge scripts support the form workflows (Table 5.4). These scripts retrieve related records, calculate booking prices, validate booking deadlines, create CRM contacts and update cancellation statuses. They are configured within the respective workflow actions in the Application IDE. The complete source code can be presented in Appendix C. Figure 5.5 shows the booking price calculation script in the Application IDE.

Table 5.4: Deluge Scripts

ID	Script	Kind	Purpose
DS-01	Auto_calculate_booking_price	Form workflow script	Retrieves the selected show's ticket price and calculates the total booking amount using the number of seats
DS-02	Block_Booking_After_Deadline	Form validation script	Compares the booking date with the selected show's booking deadline and prevents invalid submissions
DS-03	Auto_Set_Booking_Status_Confirmed	Form workflow script	Automatically assigns Confirmed status to newly created bookings
DS-04	Sync_Customer_to_Zoho_CRM	Form workflow script	Creates a Contact record in Zoho CRM using customer information
DS-05	Cancel_Booking_on_Submission	Form workflow script	Retrieves the associated booking and updates its status to Cancelled

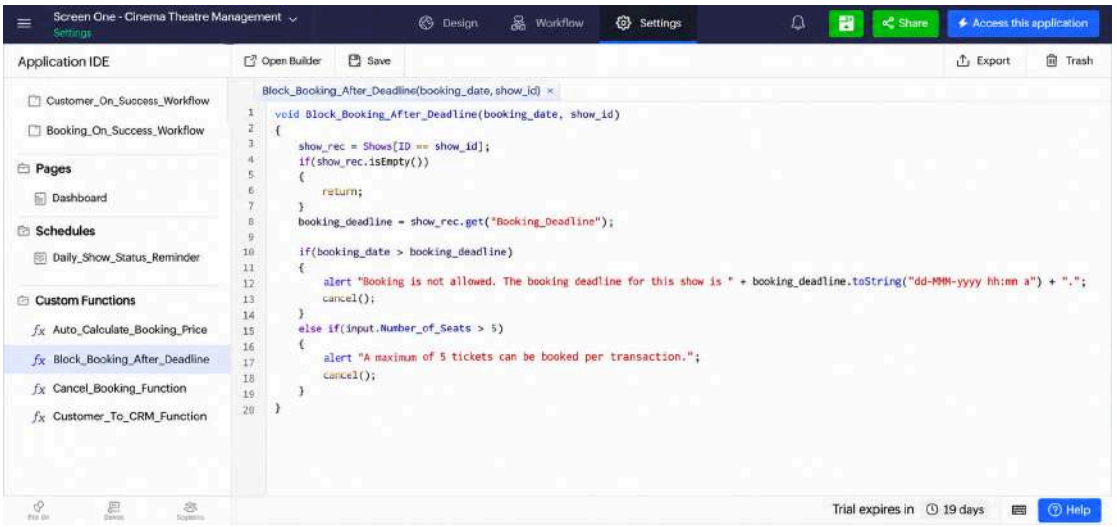


Figure 5.3: Block_Booking_After_Deadline Script in the Application IDE

5.6 Integrations

The application integrates with Zoho CRM and the AI-based OCR functionality of Zoho Creator. The Zoho CRM connection, named `screen_one_crm`, is configured with permissions to fetch, update and create Contact records. When a new Customer record is submitted, the `Sync_Customer_to_Zoho_CRM` workflow transfers the

customer's name, phone number and e-mail address to Zoho CRM and creates a corresponding Contact record.

The Movie Distribution form uses the configured AI OCR model named `Movie_Distribution_OCR`. When a distributor agreement is uploaded, the model processes the document and extracts eight details, including distributor name, agreement number, agreement amount, distribution territory, licence dates, movie title and release date. These extracted details are made available through the corresponding OCR-enabled fields in the form.

5.7 Users, Roles and Security Configuration

The application includes different permission profiles to manage access to forms and reports. These include Ticketing Agent, Guest, Show Scheduler, Content Administrator, Cinema Manager and other configured profiles, along with the application owner who acts as Administrator. The profiles provide different access levels according to their assigned responsibilities. Figure 5.6 shows the permission profiles configured in Zoho Creator. Further security and governance aspects are discussed in Chapter 7.

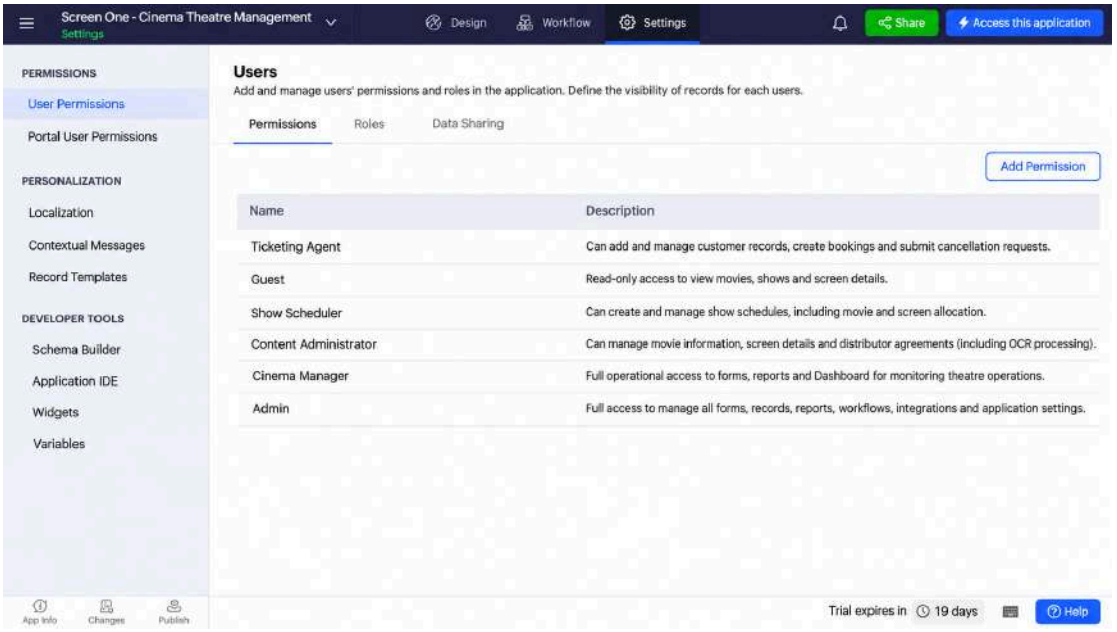


Figure 5.4: Permission Profiles in the Application

CHAPTER 6
TESTING AND DEBUGGING

6.1 Test Strategy

Testing combined configuration review and observation of the live application. Each test case was derived from a functional requirement and checked against the corresponding form, workflow, Deluge script or integration in the Application IDE. The cases cover customer registration, movie and screen management, show scheduling, booking calculations, cancellation processing, CRM synchronisation, OCR extraction, reports and access control.

6.2 Test Cases and Results

Table 6.1: Test Cases and Results

TC ID	Req.	Test Input / Action	Expected Result	Status
TC-01	FR-01	Create a customer with name, phone number and e-mail address	Customer record saved and listed in All Customers	Pass
TC-02	FR-02	Create a movie with title, genre, language and duration	Movie record saved and displayed in All Movies	Pass
TC-03	FR-03	Add a screen with screen type and seating capacity	Screen record saved and displayed in All Screens	Pass
TC-04	FR-04	Schedule a show by selecting a movie, screen, date and time	Show record saved with the selected movie and screen	Pass
TC-05	FR-05	Create a booking by selecting a customer and show	Booking record saved with the selected customer and show	Pass
TC-06	FR-06	Enter ticket quantity and select a show with a defined ticket price	Total booking amount calculated automatically	Pass
TC-07	FR-07	Enter a booking quantity exceeding five tickets	Validation message displayed and invalid booking prevented	Pass
TC-08	FR-08	Attempt to book a show after its booking deadline	Booking blocked and appropriate validation message displayed	Pass
TC-09	FR-09	Submit a cancellation request with linked booking information	Cancellation request record saved successfully	Pass

TC ID	Req.	Test Input / Action	Expected Result	Status
TC-10	FR-10	Submit a cancellation request for an existing booking	Linked booking status automatically updated to Cancelled	Pass
TC-11	FR-11	Create a movie distribution record with distributor and agreement details	Distribution record saved with the entered information	Pass
TC-12	FR-12	Upload a distributor certificate through the Movie Distribution form	Uploaded document processed using the configured OCR model	Pass
TC-13	FR-13	Upload a certificate containing agreement details	Eight specified agreement details extracted into corresponding fields	Pass
TC-14	FR-14	Submit a new customer record through Zoho Creator	Corresponding Contact record created in Zoho CRM	Pass
TC-15	FR-15	Open the application reports	Customer, movie, screen, show and booking reports displayed	Pass
TC-16	NFR-02	Open the application permission settings	Configured permission profiles and access settings displayed	Pass
TC-17	NFR-03	Open the application through a web browser	Application forms and reports displayed through the browser interface	Pass
TC-18	NFR-04	Access the application through Zoho Creator hosting	Application accessible through the hosted environment	Pass
TC-19	NFR-06	Review stored booking and cancellation records	Records available for administrative review and reference	Pass

6.3 User Acceptance Testing

Acceptance was evaluated against the main functions available in the application. The verification criteria cover customer records, movie and show management, ticket bookings, cancellation updates, CRM integration and OCR-based distributor document processing. Table 6.2 lists the acceptance criteria and the results to be recorded.

Table 6.2: Acceptance Verification from the Live Dashboard

S.No.	Acceptance Criterion	Observed Result	Accepted
1	Customer and movie information can be maintained in one application	To be recorded during live testing	Pending
2	Show schedules and screen allocations can be managed	To be recorded during live testing	Pending
3	Ticket bookings and automatic price calculations are available	To be recorded during live testing	Pending
4	Booking cancellations update the corresponding booking status	To be recorded during live testing	Pending
5	Customer information and distributor certificates can be processed through CRM and OCR integrations	To be recorded during live testing	Pending

6.4 Debugging and Defect Log

Review of the workflow definitions and application configuration identified an inactive scheduled workflow. The issue is recorded below as open and is proposed for correction before deployment. Other workflow and integration behaviours should be verified through live execution.

Table 6.3: Defect Log

Defect ID	Description	Severity	Status
D-01	The Daily_Show_Status_Reminder scheduled workflow is configured but currently inactive; therefore, automatic e-mail notifications for upcoming shows are not operational.	Low	Open

CHAPTER 7

DEPLOYMENT, OPTIMISATION, SECURITY AND GOVERNANCE

7.1 Deployment

The application is developed using Zoho Creator and configured within its development workspace. Zoho Creator provides publishing facilities through which the application can be made accessible using a live URL. The application uses the Zoho Creator Trial Edition and contains seven forms, six reports, one Dashboard page, configured workflows, a Zoho CRM connection and an AI-based OCR model. A separate sandbox-to-production promotion has not been established for this project. Table 7.1 presents the deployment checklist.

Table 7.1: Deployment Checklist

S.No.	Checklist Item	Status
1	Application created and configured in Zoho Creator	Done
2	Forms, reports and Dashboard page configured	Done
3	Booking and cancellation workflows configured	Done
4	Zoho CRM connection configured for customer synchronisation	Done
5	AI OCR model configured for distributor certificate extraction	Done
6	Application published and accessible through its live URL	Verify
7	Separate sandbox / production promotion	Not carried out
8	Edition	Trial Edition

7.2 Performance Optimisation

No load testing was carried out, so no timing figures are claimed. The design choices that support performance are listed in Table 7.2.

Table 7.2: Performance-Oriented Design Choices

Area	Choice Made	Benefit
Derived values	Total booking amount calculated automatically using ticket quantity and ticket price	Reduces manual calculations and maintains consistent booking amounts

Area	Choice Made	Benefit
Lookup relationships	Customer, Movie, Screen and Show records connected through lookup fields	Reduces duplicated information and maintains record relationships
Booking validation	Deluge workflows check booking deadlines and ticket quantity restrictions	Prevents invalid booking submissions
Dashboard	KPI tiles and charts display booking and show information	Provides a quick overview without manually compiling records
CRM integration	Customer details transferred through a configured Zoho CRM connection	Avoids repeated manual entry of customer contact information
OCR processing	AI OCR model extracts eight distributor agreement fields from uploaded documents	Reduces manual entry of distributor agreement details

7.3 Security

Table 7.3: Security Measures

Area	Measure in the Application
Authentication	Access through authenticated Zoho accounts and the Zoho Creator environment
Authorisation	Configured permission profiles provide different levels of access to application components
Customer information	Customer names, phone numbers and e-mail addresses are maintained in the Customer form
Document handling	Distributor agreements are uploaded and processed through the configured AI OCR model
Transport	Application access is provided through Zoho Creator's HTTPS-hosted environment
CRM connection	Customer information is transferred using the configured Zoho CRM connection
Secrets	CRM authentication is managed through the configured Zoho connection rather than manually storing credentials in Deluge scripts

7.4 Governance and Administration

The Zoho Creator Application IDE provides a central location for reviewing forms, reports, workflows, Deluge scripts, connections and application settings. The application owner can manage configuration changes and monitor the different components through the development environment. Consistent form names, field link names and workflow identifiers support easier maintenance and identification of application components. Zoho Creator also provides facilities such as Audit Trail, Backup, Sandbox and Logs for administration and governance. However, separate backup scheduling, audit review procedures and sandbox-based deployment have not been established as part of this project.

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Achievement of Objectives

The Staff registers a ticket booking by selecting the customer and scheduled show, then entering the required ticket quantity. The system validates the booking deadline and restricts individual bookings to a maximum of five tickets. It automatically calculates the total booking amount using the ticket quantity and ticket price and assigns the booking status as Confirmed. If a cancellation request is submitted, the system retrieves the linked booking and updates its status to Cancelled. The Administrator can monitor theatre operations, including customer records, movie schedules, bookings and cancellation details through reports and the Dashboard page.

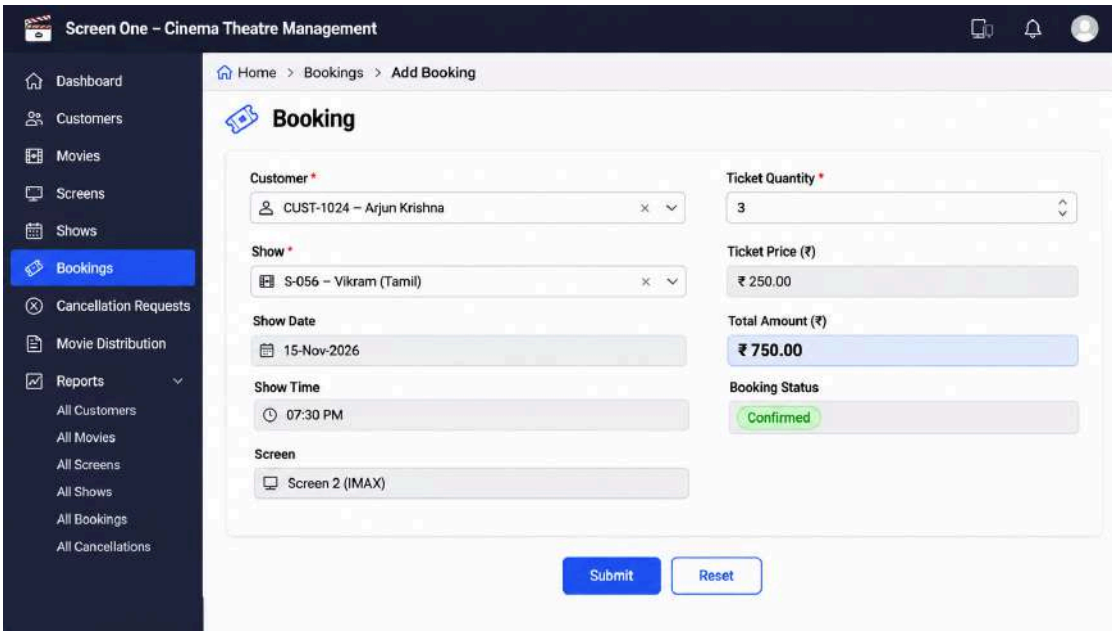


Figure 8.1: Scenario Step 1 – Customer and Ticket Booking Registration

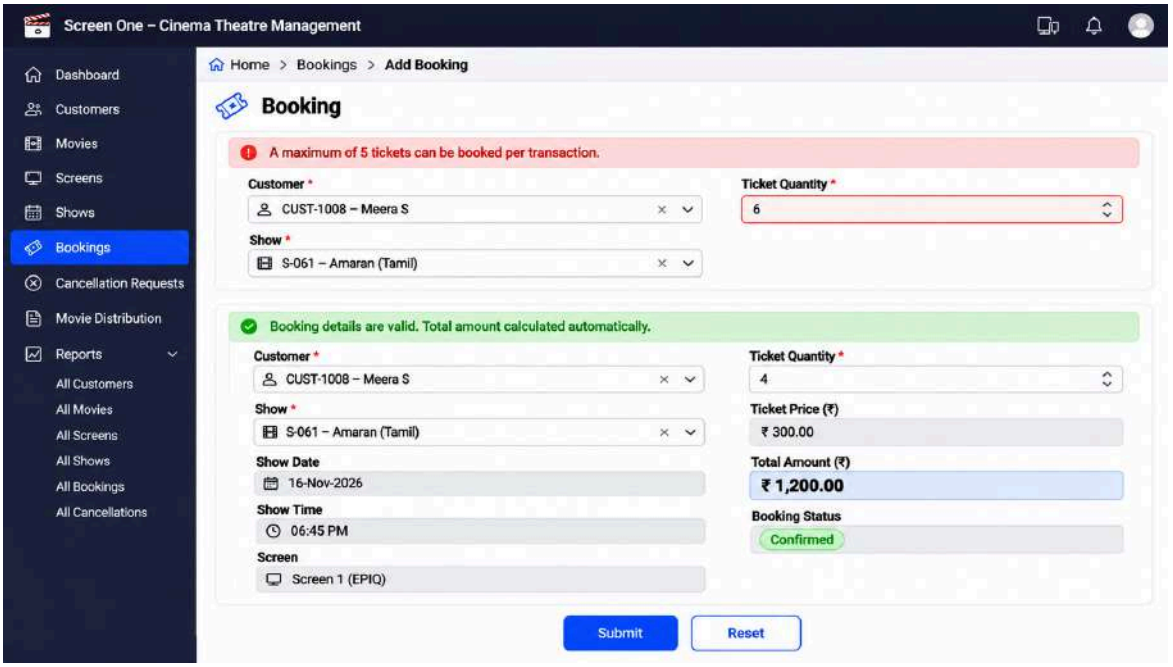


Figure 8.2: Scenario Step 2 – Automatic Booking Validation and Price Calculation

8.2 Achievement of Objectives

Table 8.1: Mapping of Objectives to Outcomes

Objective	Outcome Achieved	Evidence
Structured data model and forms	Seven forms with lookup relationships for managing cinema operations	Section 4.2, Tables 4.2–4.8
Automate booking and cancellation	Deluge workflows for price calculation, deadline validation, booking confirmation and cancellation processing	Section 5.4, Table 5.2
Dashboard and reports	Six list reports and a dashboard containing KPI tiles, charts and quick-access options	Section 5.3, Figures 5.3–5.4
Zoho CRM integration	Customer information synchronisation with Zoho CRM through a configured connection	Section 5.6
AI-based document processing	OCR model configured to extract eight distributor agreement details	Sections 4.3 and 5.6
Application security and governance	Permission profiles and Zoho Creator administration facilities available	Chapter 7, Section 7.3

8.3 Comparison with the Existing System

Table 8.2: Existing Manual Process versus the Proposed System

Aspect	Existing (Manual)	Proposed (Cinema Theatre Management System)
Record keeping	Customer, movie and booking details maintained separately	Structured records maintained through Zoho Creator forms
Movie management	Movie information recorded and updated manually	Centralised Movie form and report
Show scheduling	Screen allocation and show timings maintained manually	Show form connects movies and screens through lookup fields
Ticket booking	Booking details and ticket amounts calculated manually	Booking form with automatic price calculation
Booking validation	Ticket limits and booking deadlines checked manually	Deluge workflows validate booking conditions
Cancellation handling	Booking cancellation updated manually	Cancellation workflow automatically updates linked booking status
Distributor agreements	Agreement details entered manually from certificates	AI OCR model extracts eight distributor agreement fields
Customer management	Customer information entered separately in different records	Zoho CRM integration creates corresponding Contact records
Reports and monitoring	Information compiled manually	List reports and dashboard provide organised information

8.4 Discussion of Results

The developed application brings together the major activities involved in cinema theatre administration within a single Zoho Creator environment. The seven forms establish a structured data model, while lookup relationships connect customers, movies, screens, shows and bookings. The Zoho CRM integration reduces repeated customer data entry, while the AI-based OCR model supports the extraction of distributor agreement information from uploaded certificates. These features demonstrate the practical application of low-code development, workflow automation and AI-assisted document processing.

8.5 Limitations

The application has certain limitations that should be considered. The scheduled notification workflow is configured but currently inactive, so automatic

upcoming-show e-mail notifications are not operational. Formal load testing has not been performed, and therefore no measured performance figures are available. The application uses the Zoho Creator Trial Edition, which may impose restrictions on continued usage and available resources. The OCR extraction process may require manual verification when uploaded certificates are unclear or contain inconsistent formats. The current CRM integration focuses on creating Contact records when customer information is submitted. Advanced CRM synchronisation, such as duplicate detection and updating existing contacts, is not established. Similarly, the application does not currently include a fully implemented online customer booking portal or payment gateway integration.

8.6 Low-Code Reflection

The project demonstrates how a functional business application can be developed using a low-code platform with limited conventional programming. Zoho Creator provides visual form creation, lookup relationships, reports, dashboards and workflow configuration, reducing the amount of code required. Deluge scripting was used where additional logic was necessary, particularly for booking calculations, deadline validation, automatic status updates and CRM synchronisation. The integration of the AI-based OCR model further demonstrates how built-in platform capabilities can reduce repetitive document processing. However, the project also highlights the dependency on the platform's available features, subscription plans and integration capabilities. Overall, Zoho Creator provided a practical development environment for implementing the Cinema Theatre Management System within the project scope.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

The Cinema Theatre Management System provides a centralised solution for managing cinema operations using Zoho Creator. Seven forms, six reports and a dashboard organise customer, movie, screen, show, booking, cancellation and distributor information. Deluge workflows automate ticket price calculations, booking validation, booking confirmation and cancellation processing. Zoho CRM integration supports customer contact management, while the AI-based OCR model extracts eight important details from distributor certificates. The project demonstrates how low-code development, workflow automation and AI integration can simplify theatre administration while highlighting the possibilities and limitations of platform-based application development.

9.2 Future Enhancements

- Activate the scheduled workflow to send automatic e-mail notifications about upcoming shows to the administrator.
- Introduce an online customer portal that allows customers to browse available movies, view show timings and book tickets independently.
- Integrate a secure payment gateway to support online ticket payments and transaction tracking.
- Improve CRM synchronisation by identifying existing contacts and updating records instead of creating duplicate entries.
- Enhance OCR accuracy through document validation, manual correction options and support for different distributor certificate formats.
- Introduce advanced reporting features, including revenue analysis, movie performance comparisons and screen occupancy statistics.
- Implement automated backup scheduling and sandbox-based deployment to improve application maintenance and reliability.
- Expand the system with mobile-friendly booking facilities and automated notifications for booking confirmations and cancellations.

9.3 Learning Outcomes and Career Pathways

Building the Cinema Theatre Management System provided practical experience in designing and developing a real-world business application using a low-code platform. It improved understanding of data modelling through forms, fields, lookup relationships and record management. The project also provided hands-on experience with Deluge scripting, including conditional statements, record retrieval, calculations, validation rules and automated status updates.

The development process strengthened knowledge of event-driven workflows and business process automation. Implementing ticket booking calculations, booking deadline validation and cancellation processing provided experience in translating operational requirements into automated application logic. Designing reports and dashboards also improved understanding of data presentation and administrative monitoring.

Furthermore, integrating Zoho Creator with Zoho CRM and the AI-based OCR model provided practical exposure to application integration and AI-assisted document processing. The project helped identify the advantages and limitations of low-code platforms, particularly their rapid development capabilities, customisation options and dependency on platform-specific services. It also improved problem-solving, logical thinking, debugging and application development skills.

These skills are useful for career opportunities such as Zoho Creator Developer, Low-Code Application Developer, Business Application Developer and Low-Code Solution Consultant. The knowledge gained through this project also provides a foundation for progressing towards professional software development, where programming languages, databases, APIs, artificial intelligence and external integrations can be used to extend business applications.

REFERENCES

[1] A. C. Bock and U. Frank, “Low-Code Platform,” Business & Information Systems Engineering, vol. 63, pp. 733–740, 2021.

Link: https://doi.org/10.1007/s12599-021-00726-8

[2] Zoho Corporation, “Zoho Creator Help Documentation.” [Online]. Available: https://www.zoho.com/creator/help/

[3] D. Di Ruscio, D. Kolovos, J. de Lara, A. Pierantonio, M. Tisi, and M. Wimmer, “Low-code development and model-driven engineering: Two sides of the same coin?” Software and Systems Modeling, vol. 21, pp. 437–446, 2022.

Link: https://doi.org/10.1007/s10270-021-00970-2

[4] Zoho Corporation, “Configuring Deluge Script for Form Workflows,” Zoho Creator Help. [Online]. https://help.zoho.com/portal/en/kb/creator/developer-guide/workflows/create-and-manage-form-workflows/articles/configuring-deluge-script-for-form-workflows

[5] Zoho Corporation, “Configure Zoho CRM Integration in Workflows,” Zoho Creator Help. [Online] https://help.zoho.com/portal/en/kb/creator/developer-guide/workflows/create-and-manage-on-success-action/articles/configure-integration

[6] Zoho Corporation, “Understand OCR Model,” Zoho Creator Help. [Online]. https://help.zoho.com/portal/en/kb/creator/developer-guide/microservices/ai-modeler/articles/understand-ocr-model

[7] Zoho Corporation, “Understand OCR Field,” Zoho Creator Help. [Online]. https://help.zoho.com/portal/en/kb/creator/developer-guide/forms/add-and-manage-fields/articles/fields-ocr-index

[8] Zoho Corporation, “Understand Integration with Zoho CRM,” Zoho Creator Help. [Online] https://help.zoho.com/portal/en/kb/creator/developer-guide/workflows/create-and-manage-on-success-action/articles/understand-integration

APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

The following table records the Zoho Creator features used in the development of the Cinema Theatre Management System. "Yes" indicates that the feature has been implemented, "Partial" indicates that it has been imple mented in a limited form or is configured but not fully operational, and "No" indicates that the feature has not been implemented in the application.

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Done	Where in the Application	Report Sectio n	Figure / Code No.
I	Application created and sections/menu organised	Yes	Seven forms, six reports and one Dashboard page	5.1	Fig. 5.1
II	Forms with different field types	Yes	Text, number, date, time, currency, dropdown, radio, lookup and OCR-enabled fields	4.3, 5.2	Tables 4.2–4.8
II	Lookup field (relationship between forms)	Yes	Customer, Movie, Screen, Show, Booking and Cancellation Request	4.2, 5.2	Fig. 4.2; Table 4.1
II	Subform (parent–child data)	No	Not used	8.5	—
II	Field validations / layout rules	Yes	Booking deadline validation and maximum five-ticket restriction through Deluge scripts	5.4, 5.5	Table 5.4; Table 6.1
II	List report with filter, grouping and custom action	Partial	Six list reports with available record management actions	5.3	Table 5.1

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
II	Kanban / Calendar / Timeline / Map report	No	Not used	8.5	—
II	Pivot table / chart report	Partial	Dashboard contains booking status and distribution charts	5.3	Fig. 5.2
II	Page / dashboard built with UI Builder	Yes	Dashboard page with information panels, charts and quick-access links	5.3	Fig. 5.2
II	Deluge – form events (validate / success / field actions)	Yes	Booking calculation, deadline validation, booking confirmation, CRM synchronisation and cancellation processing	5.4, 5.5	DS-01 to DS-05; Table 5.4
II	Deluge – functions and record operations	Yes	Five embedded Deluge scripts supporting booking and cancellation operations and CRM integration	5.5	Table 5.4
II	Email / in-app notifications	Partial	Upcoming show e-mail notification workflow configured but currently inactive	5.4, 8.5	Table 5.2

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
II	Form workflow	Yes	Five form workflows for booking, validation, CRM synchronisation and cancellation processing	5.4	Table 5.2
II	Scheduled workflow	Partial	Daily_Show_Status_Reminder configured for upcoming show notifications but currently inactive	5.4, 8.5	Table 5.2
II	Approval process	No	Cancellation approval is not implemented; cancellation is processed automatically through a form workflow	5.4	Table 5.3
II	Blueprint	No	Booking Approval blueprint not implemented	5.4	Table 5.3
II	Batch / report workflow	No	Booking Processing and Show Scheduling batch workflows not implemented	5.4	Table 5.3
II	Testing and debugging evidence	Partial	Nineteen test cases recorded as Pass; acceptance verification remains pending	6.1–6.4	Tables 6.1–6.3

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
III	Integration with a Zoho service	Yes	Zoho CRM integration for automatic customer Contact creation	5.6	Table 5.2; DS-04
III	Connection / REST API call (invokeurl)	Partial	Configured Zoho CRM connection named screen_one_crm; separate invokeurl implementation not documented	5.6	—
III	Webhook or Creator REST API	No	Not used	8.5	—
III	Users, roles and permission profiles	Yes	Administrator and configured profiles including Ticketing Agent, Guest, Show Scheduler, Content Administrator and Cinema Manager	4.6, 5.7	Table 4.9; Fig. 5.4
III	Record-level access criteria / customer portal	Partial	Permission profiles configured; independent customer-facing portal not implemented	4.6, 7.3	Table 4.9; Table 7.3
III	Zia / AI feature	Yes	Movie_Distribution_OCR model extracts eight distributor agreement details	4.3, 5.6	Table 4.8

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
III	Analytics (charts / Zoho Analytics)	Partial	Dashboard KPI panels and charts; separate Zoho Analytics integration not used	5.3	Fig. 5.2
IV	Performance optimisation applied and measured	Partial	Lookup relationships, automatic calculations, validation workflows and dashboard summaries; formal load testing not conducted	7.2	Table 7.2
IV	Deployment across environments	No	Single Zoho Creator development workspace; separate sandbox-to-production promotion not established	7.1	Table 7.1
IV	Security controls (authentication, permissions and encryption)	Yes	Authenticated Zoho access, permission profiles, HTTPS hosting and configured CRM connection	7.3	Table 7.3; Fig. 5.4
IV	Governance (audit trail, backup and administration)	Partial	Zoho Creator administration facilities available; separate backup scheduling and audit review procedures not established	7.4	—

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
I	Application created and sections/menu organised	Yes	Seven forms, six reports and one Dashboard page	5.1	Fig. 5.1
II	Forms with different field types	Yes	Text, number, date, time, currency, dropdown, radio, lookup and OCR-enabled fields	4.3, 5.2	Tables 4.2–4.8

Total features implemented: 20 / 32 (6 more partially implemented).

APPENDIX B – COURSE OUTCOME MAPPING

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Ch. 1, 2	Good	
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Ch. 4, 5 (5.1–5.5), 6	Good	
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security.	Ch. 4.6, 5.6, 5.7	Satisfactory	
CO4	Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices.	Ch. 7	Satisfactory	
CO5	Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer.	Ch. 5.2, 5.5, 8.4, 8.5, 9.3	Good	

APPENDIX C – COMPLETE DELUGE SOURCE CODE

C.1 Auto_calculate_booking_price (DS-01)

```
None
// Script ID : DS-01
// Location  : Booking Form → Workflow → Auto_calculate_booking_price
// Purpose   : Automatically calculate the total booking amount using
the selected show's ticket price and ticket quantity


// Get the number of seats from user input
noOfSeats = input.no_of_seats;
// Fetch the show record to get ticket price
showRecord = show[ID == input.show];
ticketPrice = showRecord.ticket_price;
// Calculate total price
if(noOfSeats != null && ticketPrice != null)
{
	totalPrice = noOfSeats * ticketPrice;
	input.total_price = totalPrice;
}
```

C.2 Block_Booking_After_Deadline (DS-02)

```
None
// Script ID : DS-02
// Location  : Booking Form → Workflow → Block_Booking_After_Deadline
// Purpose   : Validate the booking date against the show's booking
deadline and prevent bookings after the permitted time


// Fetch the show record using the lookup field
showRecord = show[ID == input.show];
// Get the booking deadline from the show record
bookingDeadline = showRecord.booking_deadline;
// Get the booking date from input
bookingDate = input.booking_date;
// Compare booking date with booking deadline
if(bookingDate > bookingDeadline)
{
	// Display error message
	alert(booking_date,"Booking date cannot exceed the show's booking
deadline.");
	// Cancel the submission
	cancel submit;
}
```

C.3 Auto_Set_Booking_Status_Confirmed (DS-03)

```
// Script ID : DS-03
// Location  : Booking Form → Workflow → uto_Set_Booking_Status_Confirmed
// Purpose   : Automatically assign Confirmed status to newly submitted
bookings and display a success message

// Set the booking status to Confirmed when a new booking is created
input.booking_status = "Confirmed";
success message "Booking Confirmed Successfully!";
```

C.4 Sync_Customer_to_Zoho_CRM (DS-04)

```
None
// Script ID : DS-04
// Location  : Customer Form → Workflow → Sync_Customer_to_Zoho_CRM
// Purpose   : Create a corresponding Contact record in Zoho CRM
using the submitted customer information

contactMap = Map();
contactMap.put("Last_Name",input.name);
contactMap.put("Phone",input.phone_number);
contactMap.put("Email",input.email);
response =
zoho.crm.createRecord("Contacts",contactMap,Map(),"screen_one_crm");
info response;
```

C.5 Cancel_Booking_on_Submission (DS-05)

```
// Script ID : DS-05
// Location  : Cancellation Request Form → Workflow →
Cancel_Booking_on_Submission
// Purpose   : Retrieve the linked booking and automatically update its
booking status to Cancelled when a request is submitted

// Fetch the booking record using the lookup field from input
bookingRecord = booking[ID == input.booking.ID];
// Update the booking status to Cancelled
if(bookingRecord != null)
{
       bookingRecord.booking_status="Cancelled";
       // Display success message
       success message "Your booking has been cancelled successfully";
}
else
{
       // Handle case where booking record is not found
       success message "Booking record not found. Cancellation could not
be processed.";
}
```

APPENDIX D – USER MANUAL

D.1 Accessing the Application

Open the published application URL in a web browser and sign in using the authorised Zoho account. After accessing the application, the Dashboard page provides an overview of theatre operations through KPI tiles, charts and reports. The application menu provides access to the available forms and reports, including Customer, Movie, Screen, Show, Booking, Cancellation Request and Movie Distribution. The application can be accessed through a supported web browser, while the responsive layouts support different screen sizes.

D.2 Instructions for the Administrator

- Open the Customer form and register customers by entering their name, e-mail address and contact details. After successful submission, the configured workflow creates a corresponding Contact record in Zoho CRM.
- Open the Movie form and enter movie information, including title, genre, language, duration, certificate, film format and release year.
- Open the Screen form and register theatre screens by providing the screen name, screen type and seating capacity.
- Open the Show form and schedule a movie by selecting the movie and screen, entering the show date, timing, ticket price and booking deadline.
- Open the Movie Distribution form and enter distributor information or upload a distributor certificate. The configured AI-based OCR model extracts agreement amount, agreement number, distribution territory, distributor name, licence start date, licence end date, movie title and release date.
- Use the Dashboard page to monitor booking statistics, total ticket prices, show information and movie distribution charts.
- Review the available reports to view and manage customer, movie, screen, show and booking records.

D.3 Instructions for the Subscriber

- Access the Show Report to review scheduled movies, screen allocations, show dates and timings.
- Open the Booking Report to monitor customer bookings, ticket quantities, total amounts and booking statuses.
- Review cancellation requests and check the corresponding booking status.
- Use the Dashboard to monitor booking statistics and other operational information.
- Review Movie Distribution records to verify distributor and agreement information.

APPENDIX E – EVALUATION RUBRIC

(For Examiner Use Only)

Student Name	Pranav Shanmugam	Register No.	240701395
Project Title	Cinema Theatre Management System	Date	_____

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	
3	Forms, Reports & Pages	Variety of field types, lookups/subforms, report types, UI Builder dashboards	15	
4	Deluge Scripting & Workflows	Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows	15	
5	Integration, RBAC & AI	Zoho/external integrations, connections, roles & permissions, Zia/analytics	10	
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls, audit	10	
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	15	
8	Project Report	Adherence to template and format, clarity, figures, references	5	
9	Viva-Voce	Understanding of the platform, design decisions and low-code concepts	10	
	TOTAL		100	

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor
Excellent	90–100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student.

Signature of Internal Examiner

Name: _____

Signature of External Examiner

Name: _____

APPENDIX F – VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name	Pranav Shanmugam	Register No.	240701395
Department / Section	Computer Science and Engineering Section: _____	Date of Viva	_____
Application URL Verified	Yes / No	Demo Duration	_____ min

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				
6				
7				
8				

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	**100**	

Examiners' Remarks:

_____

_____

INTERNAL EXAMINER

Name: _____

EXTERNAL EXAMINER

Name: _____